

Economic Rationality under Cognitive Limitations: The Effect of Sequential Elimination

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Abstract

We study the effect of *sequential elimination*—a choice procedure whereby options are eliminated one by one until only one remains—on economic rationality. Using a tractable framework of *limited attention*, in which individuals consider only a subset of available options, we derive the hypothesis that sequential elimination improves choice consistency with preference maximization relative to direct choice from menus among individuals with limited attention. Testing this hypothesis in a controlled experiment guided by the framework, we find causal evidence of this improvement among individuals with lower cognitive ability. Evidence from attentional heterogeneity and a minimum-time constraint treatment further corroborates limited attention as a key underlying mechanism. Patterns in elimination and choice behavior are indicative of deliberate, preference-guided processing that accords with the framework.

JEL codes: C90, D81, D91, G11, I31.

Keywords: Economic rationality, revealed preference, cognitive ability, choice procedures, sequential elimination, experiment.

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1 Introduction

The standard principle of economic rationality requires individual behavior to be consistent with preference maximization. Yet cognitive constraints can hinder individuals from selecting their best options (Simon, 1957; Selten, 1990). This raises important welfare concerns, particularly for individuals with lower cognitive ability, who are often less able to bear the economic costs of choice inconsistencies.¹ Such concerns are especially relevant in high-stakes markets with large menus, such as bank loans (Honka, Hortaçsu, and Vitorino, 2017), health services (Gaynor, Propper, and Seiler, 2016), and insurance plans (Abaluck and Gruber, 2016; Abaluck and Adams-Prassl, 2021), where deviations from rational choice are widely documented. Despite seminal work establishing choice procedures as adaptations to cognitive limitations (Simon, 1955, 1976), their potential to improve economic rationality remains underexplored.

We provide the first experimental evidence that a simple choice procedure systematically improves choice consistency among individuals facing cognitive constraints. More specifically, drawing on research linking such constraints to lower attentional capacity (e.g., Cowan et al., 2006; Unsworth, 2015), we derive testable implications from a framework of *limited attention*, adapted from Masatlioglu, Nakajima, and Ozbay (2012), in which a decision maker (DM) with standard preferences considers only a subset of available options.²

Within this framework, we compare choice consistency across two procedures. The first is the *direct procedure*, in which the DM chooses directly from menus. It serves as a benchmark for decision-making under limited attention, where overlooking the best option can generate choice inconsistencies. The second is *sequential elimination*, in which the DM eliminates options one by one until only one survives. Our analysis is informed by a synthesis of evidence from marketing and psychology—and, more recently, economics—showing that both *sequential* (Besedeš et al., 2015) and *elimination-based* (e.g., Yaniv

¹See Section 6 for a review of evidence on heterogeneity in choice consistency by cognitive ability and socioeconomic characteristics.

²In this paper, preferences are defined as complete, transitive, and monotone; formal details are provided in Section 2. We abstract from other cognitive channels (such as imprecise optimization and belief distortions) to focus on limited attention, given its prominence in large-menu settings and its tractability for generating testable predictions from choice data under these preference assumptions.

and Schul, 1997, 2000; Sokolova and Krishna, 2016) procedures can mitigate choice overload and encourage deliberation (see also Section 6 for a review). Although sequential elimination appears to integrate these potential benefits, its implications for economic rationality and the underlying mechanisms have yet to be rigorously established.

We characterize sequential elimination under the *minimum attention property*, a parsimonious condition that requires consideration of at least two options whenever multiple options remain. Eye-tracking studies reveal that individuals often attend to at least two options during decision-making (Krajbich and Rangel, 2011; Reutskaja et al., 2011). Field findings corroborate this pattern by estimating the number of considered options (Honka, Hortaçsu, and Vitorino, 2017; Barseghyan, Molinari, and Thirkettle, 2021). More fundamentally, cognitive research demonstrates that adult attention spans extend beyond two objects (see, e.g., Cowan, 2001, for a review). Converging evidence from cognitive science and economics thus motivates this condition, under which we show that sequential elimination leads to choice consistency with preference maximization.

Intuitively, our theoretical result holds because, in each elimination round, a best option survives either by being overlooked or by beating the other considered options. Rather than choosing directly from an overwhelming menu, the DM makes a sequence of elimination decisions, each consistent with their preferences and feasible given their cognitive capacity. This mechanism gives rise to the hypothesis that sequential elimination improves choice consistency relative to the direct procedure among individuals with limited attention.

To test this hypothesis, we conduct a controlled experiment guided by the framework and assess economic rationality using twenty-four risky decision problems adapted from Kim et al. (2018). Each problem presents eleven distinct options in randomized order, with each option representing a lottery that yields one of two monetary prizes with equal probability. Given the simplicity of each option, the core challenge lies in considering all available options in each menu, thereby providing a setting to study whether Sequential Elimination mitigates the consequences of limited attention.

We measure economic rationality in this setting of finite choice sets using a necessary and sufficient criterion adapted from Nishimura, Ok, and Quah (2017), who develop a

general characterization of choice consistency with preference maximization. We refer to this criterion as *GARP*, as it resembles the standard Generalized Axiom of Revealed Preference (Afriat, 1967; Varian, 1982). Our primary measure of economic rationality is choice consistency, a binary indicator of whether choices satisfy GARP. To capture the extent of deviations from this benchmark, we consider count measures, including the Houtman–Maks index (HMI, Houtman and Maks, 1985) and the number of GARP violations, with higher values corresponding to greater deviations; HMI is typically regarded as reflecting choice mistakes.

In the experiment, participants are randomly assigned to either the *Direct Procedure* or *Sequential Elimination* treatments, implemented with closely matched instructions and user interfaces.³ Because limited attention is not directly observable, we use two complementary proxies for heterogeneity related to it. Our primary proxy is a score at or below the sample median on the International Cognitive Ability Resource (ICAR) test (Condon and Revelle, 2014), motivated by evidence linking cognitive ability to attentional capacity; we refer to this group as *low-IQ* participants. Our secondary proxy is a score at or above the sample median on the inattention subscale of the Adult ADHD Self-Report Scale (ASRS), a more direct measure of attentional constraints in everyday contexts; we refer to this group as *low-attention* participants.

Our main results show that Sequential Elimination substantially improves economic rationality among low-IQ participants relative to the Direct Procedure. Choice consistency rises by 15.4 percentage points, or 30.8%. HMI and the number of GARP violations decline by 30.8% and 46.0%, respectively. The improvements are even more pronounced for variants of these measures that impose first-order stochastic dominance (FSD), a stricter normative criterion in decision-making under risk (e.g., Choi et al., 2014; Polisson, Quah, and Renou, 2020). *High-IQ* participants (those scoring above the sample median), by contrast, exhibit high levels of rationality, with negligible differences across treatments. The overall-sample pattern broadly resembles that among low-IQ participants, with statistically significant effects of Sequential Elimination on several rationality measures

³See Section 3.2 for details of the experimental treatments. Throughout this paper, where initially capitalized, the terms Direct Procedure and Sequential Elimination refer to the respective experimental treatments; otherwise, they indicate their respective conceptual meanings.

but not on choice consistency. Regression estimates with a comprehensive set of controls yield qualitatively similar patterns. Collectively, these findings support our hypothesis.

Moreover, low-attention participants only partially overlap with low-IQ participants, capturing additional variation particularly informative for the hypothesis. Among low-attention participants, Sequential Elimination significantly increases choice consistency relative to the Direct Procedure and reduces HMI and FSD-HMI, while the remaining outcomes are directionally consistent. Treatment heterogeneity across attention groups closely parallels that across IQ groups, providing convergent evidence that corroborates limited attention as a key mechanism underlying the rationality improvements attributed to Sequential Elimination.

Sequential Elimination increases response time by 17.9% relative to the Direct Procedure on average, raising the possibility that its effects operate through longer deliberation. To isolate this channel, we include a third treatment, the *Minimum-Time Procedure*, which retains direct choice from the menu but imposes a minimum response time calibrated using pilot response times under Sequential Elimination. As intended, the mean response time under this treatment is 80.8% higher than under the Direct Procedure. The Minimum-Time Procedure improves economic rationality among low-IQ participants to a degree comparable to Sequential Elimination. Regression and counterfactual analyses indicate that additional response time materially contributes to the gains under the Minimum-Time Procedure, with little evidence that it accounts for those under Sequential Elimination. By comparison, the effect of Sequential Elimination is achieved with considerably less additional time, highlighting its time efficiency.

A closer inspection of the choice process further illuminates how Sequential Elimination operates. Elimination sequences reveal systematic patterns: FSD-dominated options (i.e., inferior to another option in the menu) and riskier options are eliminated earlier, while safer options survive longer. These sequences also deviate markedly from the presentation order, weighing against a perfunctory application of the procedure. Additionally, we find suggestive evidence that these elimination patterns translate into fewer selections of FSD-dominated options among low-IQ participants than under the Direct Procedure. Taken together, these results complement our main findings on

economic rationality by supporting a behavioral account of Sequential Elimination that is consistent with deliberate, preference-guided processing.

More broadly, our findings speak to the fundamental question in economics of how to reconcile observed choice inconsistencies with preference maximization. While prior work has primarily examined how choice procedures *describe* these inconsistencies, we adopt an alternative perspective by studying how procedures can *shape* economic rationality. In doing so, we show that these inconsistencies need not be viewed as entrenched features of individual behavior but can instead depend on the procedures governing choice.

The remainder of the paper is organized as follows. Section 2 develops the framework and derives the hypothesis. Section 3 describes the experimental design. Section 4 presents the main results. Section 5 provides further analysis. Section 6 situates the findings within the related literature, and Section 7 offers concluding remarks.

2 Framework

Let $X \subseteq \mathbb{R}_+^n$ be a nonempty finite set of *options*, with typical elements x , y , and z , each representing a bundle of n goods. Let $\mathcal{X}$ be the set of all nonempty subsets of X ; this is the set of *menus* with typical elements A and B . A *choice function* c assigns to each $A \in \mathcal{X}$ a unique element $c(A) \in A$, interpreted as the option chosen from A . Let $\succeq$ be a complete, transitive, and monotone *preference relation* on X .⁴

We represent limited attention by a *consideration set mapping* γ , which assigns to each menu $A \in \mathcal{X}$ a subset $\gamma(A) \subseteq A$, called its *consideration set*. The mapping γ is said to satisfy the minimum attention property if $|\gamma(A)| \geq \min\{|A|, 2\}$ for all $A \in \mathcal{X}$; that is, the DM considers at least two options unless the menu is a singleton. This property partially overlaps with the “attention floor” of two studied by Freer and Nosratabadi (2026), who find that raising the floor from one to two entails virtually no loss of descriptive power.⁵ The mapping γ is said to exhibit *full consideration* if $\gamma(A) = A$ for

⁴For $x, y \in \mathbb{R}_+^n$, we write $x \geq y$ if $x_k \geq y_k$ for all $k = 1, \dots, n$; and $x \gg y$ if $x_k > y_k$ for all $k = 1, \dots, n$. A preference relation $\succeq$ is monotone if $x \geq y$ implies $x \succeq y$ and $x \gg y$ implies $x \succ y$ but not $y \succeq x$ (also known as weak monotonicity).

⁵Their result is obtained under a specification that also imposes the attention-filter restriction

all $A \in \mathcal{X}$.

2.1 Direct Procedure

In the direct procedure, the DM chooses an option that is weakly preferred to all others in their consideration set. The following definition is adapted from [Masatlioglu, Nakajima, and Ozbay \(2012\)](#).

Definition 1. A choice function c is a *direct choice* if there exist a preference relation $\succeq$ and a consideration set mapping γ satisfying the minimum attention property such that, for all $A \in \mathcal{X}$, $c(A)$ is a $\succeq$ -best element of $\gamma(A)$. It is a *direct choice with full consideration* if it admits such a representation in which γ exhibits full consideration.

We assess economic rationality using a criterion adapted from [Nishimura, Ok, and Quah \(2017\)](#).⁶ For any $x, y \in X$, we write xR^Dy (xR^Sy , respectively) if there exist menus $A, B \in \mathcal{X}$ and an alternative $z \in A$ such that $c(A) = x$, $c(B) = y$, and $z \geq y$ ($z \gg y$, respectively). Let R denote the transitive closure of R^D ; that is, xRy if there exist an integer $k \geq 1$ and $x_0, \dots, x_k \in X$ such that $x_0 = x$, $x_k = y$, and $x_{i-1}R^Dx_i$ for every $i = 1, \dots, k$. We refer to this criterion as GARP. Formally,

Definition 2 (GARP). A choice function c is said to satisfy GARP if, for any $x, y \in X$, xRy implies that yR^Sx does not hold.

Without full consideration, direct choice can violate GARP, as the following example illustrates. Consider menus $A = \{x, y, z\}$ and $B = \{u, v, w\}$ with $z \gg u$ and $w \gg x$. Suppose that the DM's preferences are described by $z \succeq w \succeq x \succeq u \succeq v \succeq y$ and that their consideration sets are $\gamma(A) = \{x, y\}$ and $\gamma(B) = \{u, v\}$. Direct choice then yields $c(A) = x$ and $c(B) = u$, so xRu but uR^Sx ; that is, the ordered pair (x, u) constitutes a GARP violation. With full consideration, direct choice becomes standard preference maximization, yielding $\tilde{c}(A) = z$ and $\tilde{c}(B) = w$ and satisfying GARP. More generally, limited consideration can generate GARP violations that are absent under full

of [Masatlioglu, Nakajima, and Ozbay \(2012\)](#), which we do not require.

⁶Our setting involves finite choice menus rather than classical consumption spaces. Theorem 1 of [Nishimura, Ok, and Quah \(2017\)](#), which extends [Afriat \(1967\)](#), applies to any preference relation and choice function, imposing no domain restrictions. Theorem 2 of [Cosaert and Demuynck \(2015\)](#) also shows that this adapted criterion, in their formulation, is necessary and sufficient for choice consistency with utility maximization.

consideration by causing the DM to overlook the most preferred option. The following remark formalizes these observations and underlies our hypothesis.

Remark 1. *Let c and $\tilde{c}$ be direct choices. Then the following statements hold:*

- (i) *c does not necessarily satisfy GARP.*
- (ii) *c satisfies GARP if and only if c is a direct choice with full consideration.*
- (iii) *Consequently, if c and $\tilde{c}$ share the same preference relation $\succeq$, with consideration mappings γ and $\tilde{\gamma}$ satisfying $\gamma(A) \subseteq \tilde{\gamma}(A) = A$ for all $A \in \mathcal{X}$, then the number of GARP violations under c is weakly greater than under $\tilde{c}$.*

2.2 Sequential Elimination

Remark 1 motivates an alternative procedure that achieves choice consistency without full consideration. We propose sequential elimination, in which the DM eliminates options one by one until only one survives as the choice. To illustrate, suppose again that the DM faces menu A . Reaching a final choice requires two elimination rounds. In the first round, the DM eliminates $e_1(A) = y$. In the second round, facing the remaining menu $A \setminus \{y\} = \{x, z\}$, the DM eliminates $e_2(A) = x$, leaving z as the final choice.

Formally, an elimination function e assigns to every $A \in \mathcal{X}$ a sequence $e(A) = (e_1(A), \dots, e_{|A|}(A)) \in X^{|A|}$ such that $\bigcup_{r=1}^{|A|} \{e_r(A)\} = A$. The sequence $e(A)$ fully describes the DM's elimination behavior. When $|A| \geq 2$, the DM sequentially eliminates $e_1(A), \dots, e_{|A|-1}(A)$ and ultimately selects $e_{|A|}(A)$; when $|A| = 1$, the DM bypasses elimination and chooses the sole element. For all $A \in \mathcal{X}$ and $r = 1, \dots, |A|$, let $A_r = \bigcup_{s=r}^{|A|} \{e_s(A)\}$ denote the remaining menu at round r . We model sequential elimination as follows.

Definition 3. A choice function c is a *choice by sequential elimination* if there exist a preference relation $\succeq$, a consideration set mapping γ satisfying the minimum attention property, and an elimination function e such that, for every $A \in \mathcal{X}$:

- (i) $e_r(A) \in \gamma(A_r)$ for every $r = 1, \dots, |A|$;
- (ii) $\{x \in \gamma(A_r) \mid x \succeq e_r(A), x \neq e_r(A)\} \neq \emptyset$ for every positive integer $r < |A|$;
- (iii) $e_{|A|}(A) = c(A)$.

Conditions (i) and (ii) of Definition 3 require the DM to eliminate an option from

their consideration set only if another considered option is weakly preferred to it. The minimum attention property ensures that, whenever multiple options remain, the DM considers at least two of them, so that each elimination is grounded in a binary comparison according to the DM's preferences. This rules out singleton consideration sets while other options remain, which could otherwise permit the arbitrary elimination of the best option. Condition (iii) links an elimination data set to a choice data set by designating the last remaining option as the choice.

The following proposition characterizes choice behavior under sequential elimination. The proof is in Appendix A.1.

Proposition 1. *Let c be a choice function. c satisfies GARP if and only if c is a choice by sequential elimination.*

The “if” direction of the proposition establishes that choices under sequential elimination are consistent with preference maximization.⁷ Thanks to the minimum attention property, a best option in a menu survives each elimination round in one of two ways. Either the DM overlooks it, or the DM considers it, in which case it beats every other option in the consideration set. Sequential elimination thus mitigates limited attention by decomposing a demanding preference-maximization problem into a sequence of elimination subproblems, each manageable within the DM's cognitive constraints. This also implies an attenuated role for cognitive heterogeneity in shaping choice consistency under sequential elimination.

2.3 Testable implication

Remark 1 and Proposition 1 imply that both procedures yield choice consistency under full consideration. As consideration sets contract, the scope for sequential elimination to improve choice consistency relative to the direct procedure expands, an implication particularly relevant for individuals with limited attention. Accordingly, we formulate the following hypothesis.

⁷The “only if” direction implies that consistent choice data alone cannot uniquely identify the underlying choice procedures.

Hypothesis 1. *Sequential elimination improves choice consistency relative to the direct procedure among individuals with limited attention.*

Empirically, we characterize heterogeneity relevant to limited attention using IQ and ASRS inattention as proxies, as discussed in Sections 3.3 and 4.1. The procedural comparison illuminates the sources of inconsistency. If the procedural difference aligns with the hypothesis, particularly among individuals facing greater cognitive constraints, it provides evidence for the limited-attention mechanism described in the framework.

If no such difference is detected, the interpretation depends on the level of choice consistency. Similarly low consistency across procedures would suggest alternative sources of inconsistency, such as non-standard preferences, including incompleteness (Ok, 2002) and intransitivity (Mandler, 2005), departures from the minimum attention property, or socioeconomic heterogeneity (e.g., Choi et al., 2014; Echenique, Imai, and Saito, 2023). For example, with incomplete preferences, sequential elimination need not resolve indeterminacy in binary comparisons, and thus the resulting choices may remain inconsistent. Conversely, uniformly high consistency across procedures could arise if most participants have sufficient cognitive capacity to exercise full consideration under the Direct Procedure, leaving limited scope for improvement.

3 Experimental design

Our experiment proceeds as follows. Participants are randomly assigned to one of three treatments: Direct Procedure, Sequential Elimination, or Minimum-Time Procedure. They first make economic decisions under their assigned treatment and then complete a set of cognitive tests. The experiment concludes with a survey eliciting attitudes toward inconsistency, the ASRS, and demographic information (age, gender, and education). Details of the design are provided below, with instructions and screenshots in Appendix B.1.

3.1 Rationality measures

Our experiment features twenty-five risky decision problems, of which twenty-four are adapted from [Kim et al. \(2018\)](#) to measure economic rationality and one is embedded as a comprehension check.⁸ Each problem consists of eleven distinct options presented in randomized order. Every option, denoted as $[x_1, x_2]$, yields either x_1 or x_2 tokens with equal probability. The twenty-four rationality problems are derived from unique budget lines, each characterized by a specific price–endowment combination. Guided by our framework, the design keeps any given binary comparison highly tractable while making exhaustive consideration of available options the main cognitive bottleneck. The decision problems are listed in Appendix Table [B.3](#); graphical representations of a decision problem and a GARP violation in two-dimensional space appear in Appendix Figure [B.7](#).

Our principal measure, choice consistency, equals one if a participant’s choices are consistent with preference maximization (i.e., satisfy GARP) and zero otherwise. We employ HMI to capture the extent of inconsistency, defined as the minimum number of observations that must be removed to restore consistency among the remaining choices; the removed observations are typically interpreted as choice mistakes.

For robustness, we consider the number of GARP violations, where higher values indicate greater deviations from rationality. We further construct FSD-compliant measures—FSD-consistency, FSD-HMI, and the number of FSD-GARP violations—that impose first-order stochastic dominance, a regularity condition stricter than monotonicity, while remaining otherwise identical in construction and interpretation to their monotonicity-only counterparts (see Appendix [A.2](#) for formal definitions). These additional measures supplement choice consistency and HMI, varying in granularity and stringency.

⁸The embedded comprehension check presents eleven strictly increasing options ranging from $[0, 0]$ to $[111, 111]$. Choosing any option other than the strictly dominant $[111, 111]$ indicates a lack of comprehension.

3.2 Treatment variations

All treatments share nearly identical interfaces and instructions and differ only in the imposed choice procedure. In each decision problem, participants choose an option from a vertical list on the left side of the screen. Each treatment begins with a practice trial.

Direct Procedure. Participants make their choice after sequentially examining all options.⁹ They mark an option as examined by clicking on it, which moves it to a *Choice List* on the right side of the screen. After all options have been moved to the Choice List, participants select their preferred option from that list.

Sequential Elimination. Participants make their choice by sequentially eliminating options they do not prefer until only one remains. They eliminate an option by clicking on it, which moves it to a *Trash* box on the right side of the screen. To reduce errors from trembles or unfamiliarity, eliminated options can be reinstated at any time before submission.

Minimum-Time Procedure. Participants make their choice from the initial list of options and must spend at least 35 seconds on each decision problem before submission. This time floor is calibrated from pilot data to approximate the average response time under Sequential Elimination (33.66 seconds). A timer on the right side of the screen displays elapsed time for each problem.

Although alternative implementations are possible, this controlled design helps isolate the mechanisms through which choice procedures shape choice behavior. The Direct Procedure requires a click on every option before the final choice, thereby serving as a benchmark with exposure to the full menu of options comparable to Sequential Elimination. This limits early search stopping due to satisficing motives such as effort reduction or regret avoidance (Simon, 1955; Caplin, Dean, and Martin, 2011) by preventing participants from submitting a choice after interacting with only part of the menu. Contrasting the two treatments therefore speaks to whether the effect of Sequential Elimination operates through mitigating the impact of cognitive limitations in processing the full menu.

⁹To minimize potential naming effects, the Direct Procedure is introduced as *Sequential Examination* in the experiment.

3.3 Additional measures

We measure cognitive ability using the IQ score from the ICAR test, which comprises five matrix-reasoning and five three-dimensional rotation tasks that primarily capture analytical reasoning and problem-solving ability. IQ serves as a broad proxy for cognitive capacity, which is assumed to be positively associated with the size of the consideration set given its relation to attention and working memory (Unsworth et al., 2014). We also administer the self-reported ASRS (v1.1), which comprises two nine-item subscales for inattention and hyperactivity-impulsivity, with higher scores indicating greater levels of each. The inattention subscale captures everyday patterns such as difficulty sustaining attention, distractibility, and trouble starting or completing tasks. It therefore complements IQ by providing a more direct measure of attentional constraints. Together with Remark 1(iii), the interpretation of these proxies implies that participants with lower IQ or higher ASRS inattention scores are expected to exhibit lower choice consistency under the Direct Procedure.

Selective attention and working memory are included as controls, measured via the Stroop task (Stroop, 1935) and the Sternberg task (Sternberg, 1966), respectively.¹⁰ Finally, we elicit attitudes toward inconsistency using a decision scenario illustrating the attraction effect, a canonical form of context-dependent inconsistency (Huber, Payne, and Puto, 1982). Participants rate how at ease they are with the inconsistency on a 0–10 scale, with higher values indicating weaker aversion. We include this rating as a control for potential non-standard preferences.

3.4 Implementation and sample

The experiment was conducted online via Qualtrics in December 2025. A pilot study informed the power calculations and minimum detectable effects reported in the pre-registration. Participants were recruited from Prolific and restricted to those with

¹⁰Selective attention refers to the ability to focus on relevant information while suppressing distractions. In the Stroop task, participants identify the print color of incongruent words, requiring inhibition of automatic responses. Working memory refers to the capacity for temporary storage and maintenance of information. In the Sternberg task, participants memorize short number sequences and indicate whether a subsequently presented number was included in the sequence. Scores are based on the number of correct responses in each task.

an approval rating of at least 95% and a minimum of 10 prior submissions. Upon completion, participants received a participation fee of £1 and an additional payment of up to £14.60, contingent on their choices in the decision problems and performance in the cognitive tests. On average, participants completed the experiment in 42 minutes and earned £8.14.

Our sample comprises 616 participants (52.3% female), with 201–212 observations per treatment.¹¹ By design, individual characteristics—including cognitive measures, the ASRS subscales, attitudes toward inconsistency, and demographics—are balanced across treatment groups. The mean age is approximately 44 years. Nearly all participants (99.5%) had completed at least secondary education, and 69.0% had attained an undergraduate degree or higher at the time of the experiment. Descriptive statistics and balance checks are reported in Appendix Table C.1, with corresponding histograms in Appendix Figure C.1.

4 Main results

4.1 Empirical proxies and participant classification

Before analyzing treatment effects, we establish the empirical relevance of the cognitive and attentional measures, with the correlation matrix reported in Appendix Figure D.1. IQ emerges as the most robust predictor of economic rationality, with significant correlations across our measures (e.g., Pearson $r = 0.11$ with choice consistency). Although selective attention and working memory are positively correlated with IQ, they are only occasionally correlated with the rationality measures, plausibly reflecting their more domain-specific functions. ASRS inattention exhibits a weak correlation with IQ and the rationality measures, despite a significant one with selective attention.

As pre-registered, we test Hypothesis 1 using low-IQ participants as the primary

¹¹A total of 662 participants (52.7% female; 218–225 per treatment) completed the experiment. During the initial instructions, participants were required to pass three comprehension questions about the decision task, each within two attempts. Forty-six participants who failed the embedded comprehension check were excluded from the analysis; of these, two also failed the instruction-following check embedded in the ASRS screen (see Appendix Table B.2) and were ineligible for compensation, as specified in the initial instructions.

proxy and low-attention participants as a complementary proxy for individuals with limited attention. Low-IQ participants score at or below the sample median, while high-IQ participants score above it.¹² We analogously classify participants with ASRS inattention scores at or above the sample median as low-attention and those below the median as high-attention. The composition of these groups is reported in Appendix Table C.2: the low-IQ and low-attention groups comprise 53.9% and 57.1% of the sample, respectively (Panels A and B), and 53.7% of low-attention participants are also low-IQ (Panel C), indicating that the ASRS-based classification captures partly distinct variation from the IQ-based split.

4.2 Treatment effects by IQ

Figure 1 reports mean rationality measures by treatment and IQ group. We examine treatment differences using chi-square tests for binary outcomes and Mann–Whitney U tests for count outcomes. Corresponding p -values for pairwise treatment comparisons are provided in Appendix Table D.1, and those for pairwise comparisons between IQ groups within each treatment in Appendix Table D.2.

The top-left panel shows that Sequential Elimination significantly increases choice consistency among low-IQ participants by 15.4 percentage points, or 30.8%, relative to the Direct Procedure (0.654 vs. 0.500, $p = 0.020$). HMI and GARP violations also decline by 30.8% (0.692 vs. 1.000, $p = 0.021$) and 46.0% (7.570 vs. 14.017, $p = 0.011$), respectively, in the left-middle and left-bottom panels.

The right panels illustrate more pronounced effects for the FSD-compliant measures: FSD-consistency rises by 16.4 percentage points, or 50.9% (0.486 vs. 0.322, $p = 0.012$), while FSD-HMI and FSD-GARP violations fall by 38.0% (1.393 vs. 2.246, $p = 0.003$) and 50.0% (27.561 vs. 55.127, $p = 0.001$), respectively. The empirical cumulative distributions of the count measures display corresponding shifts from the Direct Procedure to Sequential Elimination (Appendix Figure D.2). Taken together, these treatment effects provide causal evidence of improvements in economic rationality among low-IQ

¹²The sample's IQ scores range from 0 to 9, with the first and third quartiles at 3 and 6, respectively. The mean IQ is approximately 4.51, with a median of 4 and a standard deviation of about 2.14. Thus, splitting the sample by either the mean or the median yields identical low-IQ and high-IQ groups.

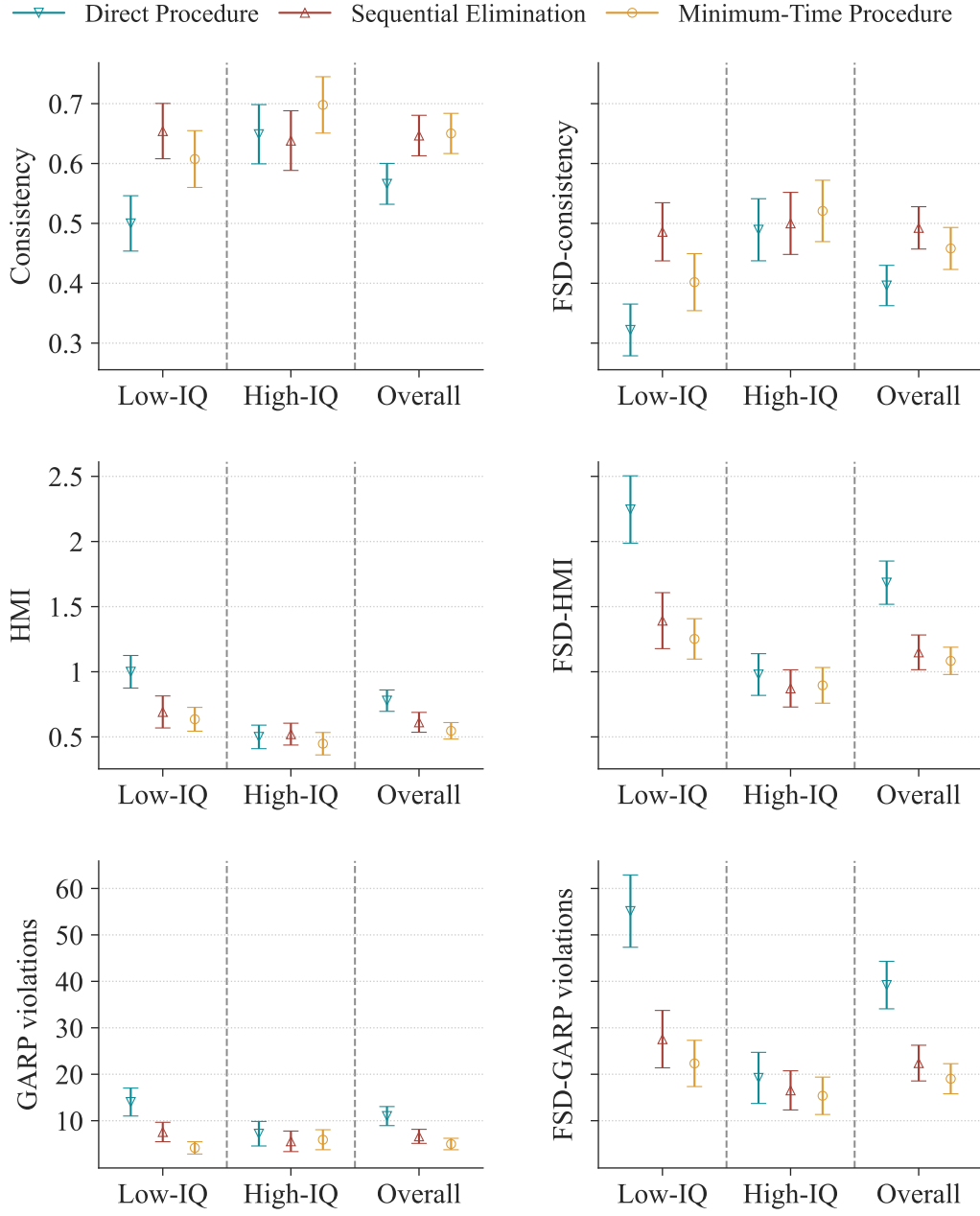


FIGURE 1. Economic rationality across treatments by IQ group. Mean values of each rationality measure are shown for low-IQ participants, high-IQ participants, and the overall sample in each treatment, with error bars representing standard errors.

participants, thereby supporting Hypothesis 1.

High-IQ participants exhibit no statistically significant treatment differences on any measure, as implied by the framework (Remark 1(ii) and Proposition 1). Their economic rationality is higher than that of low-IQ participants under the Direct Procedure

(e.g., 0.649 vs. 0.500 in choice consistency, $p = 0.030$), in line with prior studies and Remark 1(iii). Under Sequential Elimination, this gap largely disappears (e.g., 0.638 vs. 0.654 in choice consistency, $p = 0.814$), consistent with the attenuated role of cognitive heterogeneity discussed in Proposition 1.

In the overall sample, Sequential Elimination raises choice consistency by 8.1 percentage points relative to the Direct Procedure (0.647 vs. 0.566, $p = 0.093$) and reduces HMI (0.612 vs. 0.778, $p = 0.095$). Statistically significant effects emerge for GARP violations and all FSD-compliant measures, notably FSD-consistency (0.493 vs. 0.396, $p = 0.049$).

Turning to the Minimum-Time Procedure, we find improvements among low-IQ participants that are comparable in magnitude to those under Sequential Elimination. Choice consistency and FSD-consistency increase by 10.7 and 8.0 percentage points, respectively, corresponding to gains of 21.4% (0.607 vs. 0.500, $p = 0.106$) and 24.8% (0.402 vs. 0.322, $p = 0.213$), although these differences do not reach statistical significance. Clearer evidence emerges among the intensive-margin measures: HMI and FSD-HMI decrease by 36.4% (0.636 vs. 1.000, $p = 0.052$) and 44.3% (1.252 vs. 2.246, $p = 0.016$), respectively, while GARP and FSD-GARP violations fall by 70.4% and 59.5%, respectively (4.150 vs. 14.017, $p = 0.014$; 22.327 vs. 55.127, $p = 0.003$). Treatment differences remain negligible among high-IQ participants.¹³

Pairwise comparisons between Sequential Elimination and the Minimum-Time Procedure reveal no statistically significant differences within either IQ group or in the overall sample. Persistent deviations from rationality across IQ groups and treatments nevertheless point to potential sources of inconsistency beyond limited attention and the choice procedures examined here.

4.2.1 Regression estimates

As a robustness check, we estimate regressions controlling for cognitive measures, ASRS inattention, attitudes toward inconsistency, and demographics. Table 1 reports average marginal effects to facilitate interpretation, with the corresponding regression

¹³The gap in rationality between high-IQ and low-IQ participants also narrows under the Minimum-Time Procedure, although it remains statistically significant for some measures (Appendix Table D.2).

coefficients reported in Appendix Table D.3. Columns 1 and 4 use logistic regressions for the binary measures, while the remaining columns use negative binomial regressions for the count measures.¹⁴ The estimates closely align with the nonparametric evidence in direction and magnitude, corroborating Hypothesis 1.

Among low-IQ participants, the estimated effect of Sequential Elimination on choice consistency is 14.4 percentage points ($se = 0.065$; Column 1). The estimated reductions in HMI and GARP violations are marginally significant, at -0.292 ($se = 0.172$) and -7.483 ($se = 3.964$), respectively. Effects on all FSD-compliant measures remain statistically significant and larger than their non-FSD counterparts (Columns 4–6). A similar pattern emerges for the Minimum-Time Procedure: the estimated effect on choice consistency is 11.0 percentage points ($se = 0.066$), approaching conventional significance levels, while the count-measure reductions attain statistical significance. Treatment effects among high-IQ participants remain statistically nonsignificant.

These estimates reinforce the evidence that both treatments lead to economically meaningful improvements among low-IQ participants. Moreover, Sequential Elimination reveals sizable effects in the overall sample, particularly for FSD-compliant measures, with these patterns largely persisting in regressions.¹⁵ These findings leave open whether the treatment effects are confined to participants with lower IQ, prompting our subsequent analysis based on ASRS inattention.

4.3 Treatment effects by attention

We next examine whether the treatment effects are also present among low-attention participants. For brevity, we discuss the average marginal effects in Table 2; the corresponding regression coefficients appear in Appendix Table D.6. The descriptive statistics and nonparametric tests show broadly similar patterns to the regression results, as reported in Appendix Figures D.3 and D.4 and Appendix Table D.5.

¹⁴Negative binomial regressions are suitable for count data with skewed and overdispersed distributions (Cameron and Trivedi, 2013). We use the NB2 specification, under which $\text{Var}(y \mid x) = \mu + \alpha\mu^2$, where μ is the conditional mean and α is the dispersion parameter. This specification allows the conditional variance to exceed the conditional mean.

¹⁵Interaction effects between the treatments and either the high-IQ indicator or the IQ score are generally nonsignificant across outcomes (Appendix Tables D.3 and D.4).

TABLE 1. Average marginal effects of choice procedures on economic rationality by IQ group.

	Consistency (1)	HMI (2)	GARP violations (3)	FSD-compliant		
				Consistency (4)	HMI (5)	GARP violations (6)
<i>Sequential Elimination</i>						
Low-IQ participants	0.144 (0.065)	-0.292 (0.172)	-7.483 (3.964)	0.164 (0.065)	-0.830 (0.324)	-30.549 (10.653)
High-IQ participants	0.020 (0.069)	-0.027 (0.127)	-1.726 (3.234)	0.032 (0.073)	-0.135 (0.205)	-2.325 (6.595)
Overall	0.087 (0.048)	-0.170 (0.110)	-4.829 (2.579)	0.103 (0.049)	-0.510 (0.198)	-17.536 (6.520)
<i>Minimum-Time Procedure</i>						
Low-IQ participants	0.110 (0.066)	-0.352 (0.150)	-10.875 (3.660)	0.079 (0.064)	-0.963 (0.293)	-34.189 (10.310)
High-IQ participants	0.053 (0.068)	-0.053 (0.132)	-0.359 (3.287)	0.034 (0.072)	-0.068 (0.211)	-2.668 (6.552)
Overall	0.084 (0.048)	-0.214 (0.102)	-6.027 (2.482)	0.058 (0.048)	-0.550 (0.186)	-19.657 (6.310)
Observations	616	616	616	616	616	616

Note: This table reports the average marginal effects of Sequential Elimination and the Minimum-Time Procedure on economic rationality relative to the Direct Procedure by IQ group. Columns 1 and 4 present average marginal effects from logistic regressions, while the remaining columns present those from negative binomial regressions. All specifications include a treatment–High-IQ interaction, controls for ASRS inattention, selective attention, working memory, attitude toward inconsistency, age, gender, education, and a constant. Robust standard errors are reported in parentheses.

Among low-attention participants, Sequential Elimination yields higher levels of economic rationality, most notably raising choice consistency by 19.5 percentage points ($se = 0.064$). HMI and FSD-HMI also decline significantly, with the remaining outcomes moving in consistent directions and approaching statistical significance. The Minimum-Time Procedure also increases choice consistency by 16.8 percentage points ($se = 0.065$) and markedly reduces HMI. Among high-attention participants, treatment effects are less systematic. Sequential Elimination nevertheless shows marginally significant improvements in the FSD-compliant measures, while the Minimum-Time Procedure significantly reduces FSD-HMI, possibly reflecting the group’s nontrivial share of low-IQ participants.

TABLE 2. Average marginal effects of choice procedures on economic rationality by attention group.

	FSD-compliant					
	Consistency (1)	HMI (2)	GARP violations (3)	Consistency (4)	HMI (5)	GARP violations (6)
<i>Sequential Elimination</i>						
Low-attention participants	0.195 (0.064)	-0.403 (0.144)	-5.054 (3.837)	0.092 (0.065)	-0.506 (0.250)	-15.691 (8.218)
High-attention participants	-0.039 (0.070)	0.100 (0.167)	-3.841 (2.792)	0.122 (0.072)	-0.514 (0.282)	-14.244 (8.568)
Overall	0.095 (0.047)	-0.187 (0.110)	-4.535 (2.484)	0.105 (0.048)	-0.510 (0.189)	-15.071 (5.913)
<i>Minimum-Time Procedure</i>						
Low-attention participants	0.168 (0.065)	-0.314 (0.151)	-5.216 (3.885)	0.027 (0.064)	-0.410 (0.256)	-12.405 (9.515)
High-attention participants	-0.005 (0.070)	-0.130 (0.139)	-4.291 (2.968)	0.117 (0.073)	-0.657 (0.264)	-16.019 (8.497)
Overall	0.094 (0.048)	-0.235 (0.104)	-4.820 (2.457)	0.066 (0.048)	-0.516 (0.186)	-13.954 (6.258)
Observations	616	616	616	616	616	616

Note: This table reports the average marginal effects of Sequential Elimination and the Minimum-Time Procedure on economic rationality relative to the Direct Procedure by attention group. Columns 1 and 4 present average marginal effects from logistic regressions, while the remaining columns present those from negative binomial regressions. All specifications include a treatment–high-attention interaction, controls for IQ, selective attention, working memory, attitude toward inconsistency, age, gender, education, and a constant. Robust standard errors are reported in parentheses.

In sum, treatment heterogeneity across attention and IQ measures is qualitatively and quantitatively similar, strengthening the evidence for Hypothesis 1 and for limited attention as a key underlying mechanism.

5 Further analysis

This section presents exploratory analyses that shed further light on the mechanisms underlying Sequential Elimination and its broader implications. We first investigate the role of response time in accounting for the treatment effects, given the observed improvements under the Minimum-Time Procedure. We then probe whether eliminations are preference-guided, as specified by the framework, and whether these patterns carry over to final choices. Finally, we draw on pilot evidence to explore who adopts Sequential

Elimination and whether its effects on rationality persist under voluntary adoption.

5.1 Response time

Relative to the Direct Procedure (22.4 seconds), mean response time is 17.9% and 80.8% higher under Sequential Elimination (26.4 seconds) and the Minimum-Time Procedure (40.5 seconds), respectively; this pattern is qualitatively similar across IQ groups.¹⁶ Longer response times are often associated with greater decision accuracy (see, e.g., Heitz, 2014, for a review).¹⁷ We therefore examine whether the additional response time under Sequential Elimination can account for its rationality gains. As shown in Table 3, adding response time to our main specifications reveals that longer response times are significantly associated with several count measures, e.g., HMI and FSD-HMI, though not with choice consistency. Crucially, controlling for response time materially attenuates the estimated effects of the Minimum-Time Procedure but leaves those of Sequential Elimination largely intact, underscoring the role of the latter’s procedural structure.

We also conduct two counterfactual exercises for low-IQ participants under the Direct Procedure, using the estimated response-time relationships from our regression specifications to quantify the extent to which additional time might account for the rationality gains under Sequential Elimination, as reported in Appendix Table D.11.¹⁸ First, we replace their observed response times with the response-time distribution of low-IQ participants under Sequential Elimination, holding all other covariates fixed. This results in little change in predicted rationality outcomes. Second, we calculate the additional response time implied by the estimated models to reproduce the observed mean outcomes among low-IQ participants under Sequential Elimination. The average

¹⁶Appendix Table D.9 reports response-time means by treatment and IQ group, and Appendix Table D.10 reports the corresponding regression estimates. High-IQ participants take longer than low-IQ participants under the Direct Procedure and Sequential Elimination, whereas this difference disappears under the Minimum-Time Procedure.

¹⁷This association may not hold, for example, when intuitive reasoning generates choices that are both rapid and accurate (Rubinstein, 2007, 2013).

¹⁸This analysis is based on specifications with a triple interaction among response time, treatment, and IQ group. Average marginal effects of response time by treatment and IQ group across rationality measures are reported in Appendix Table D.12, where statistically significant associations are mostly confined to high-IQ participants under Sequential Elimination, suggesting that the relationship between response time and rationality may vary by IQ group and treatment condition.

TABLE 3. Average marginal effects of response time and choice procedures on economic rationality.

	Consistency (1)	HMI (2)	GARP violations (3)	FSD-compliant		
				Consistency (4)	HMI (5)	GARP violations (6)
Response time	0.001 (0.002)	-0.008 (0.004)	-0.184 (0.070)	0.003 (0.002)	-0.021 (0.007)	-0.281 (0.212)
<i>Sequential Elimination</i>						
Low-IQ participants	0.139 (0.065)	-0.248 (0.163)	-6.402 (3.331)	0.156 (0.066)	-0.692 (0.302)	-28.454 (10.169)
High-IQ participants	0.014 (0.070)	-0.000 (0.117)	-2.067 (2.767)	0.013 (0.073)	-0.098 (0.193)	-2.965 (6.435)
Overall	0.081 (0.048)	-0.134 (0.105)	-4.403 (2.196)	0.090 (0.049)	-0.418 (0.189)	-16.703 (6.309)
<i>Minimum-Time Procedure</i>						
Low-IQ participants	0.084 (0.077)	-0.148 (0.186)	-6.682 (3.644)	0.014 (0.072)	-0.438 (0.360)	-26.949 (12.036)
High-IQ participants	0.034 (0.073)	0.054 (0.144)	1.953 (3.786)	-0.021 (0.077)	0.170 (0.253)	-0.839 (7.239)
Overall	0.061 (0.058)	-0.055 (0.131)	-2.701 (2.760)	-0.002 (0.057)	-0.158 (0.254)	-14.911 (7.761)

Note: This table reports the average marginal effects of response time, Sequential Elimination, and the Minimum-Time Procedure on economic rationality. Response time is averaged across decision problems at the individual level. Treatment effects are reported by IQ group. Columns 1 and 4 present average marginal effects from logistic regressions, while the remaining columns present those from negative binomial regressions. All specifications include controls for ASRS inattention, selective attention, working memory, attitude toward inconsistency, age, gender, education, and a constant. Robust standard errors are reported in parentheses.

implied response-time increases range from 32.7 seconds for GARP violations to 268.2 seconds for FSD-consistency, far exceeding the observed response-time gaps between the Direct Procedure and the other treatments.

Further, we decompose response time into *time to last click* and *time after last click* to help distinguish deliberation from procedural constraints.¹⁹ The former accounts for only about 40% of response time under the Minimum-Time Procedure, but closely tracks it elsewhere. Substituting these components for response time in the main

¹⁹Time to last click denotes the interval until the last click on an option, and time after last click the interval from that point until submission. See Appendix Tables D.9 and D.10 for means and regressions, respectively.

specifications yields broadly similar associations, though estimates vary across outcomes (e.g., FSD-consistency is significantly associated only with time after last click; Appendix Table D.13). One possible interpretation is that participants continue deliberating after the last click under the Minimum-Time Procedure, potentially allowing them to resolve residual uncertainty about their preferences (Ergin and Sarver, 2010). Yet this channel appears less relevant under Sequential Elimination, given the negligible time elapsed after the last click.

These analyses jointly indicate that the effect of Sequential Elimination on rationality is not driven primarily by extended deliberation. Moreover, despite comparable improvements in rationality, Sequential Elimination entails a considerably smaller increase in response time than the Minimum-Time Procedure, pointing to a more favorable trade-off between rationality and time.

5.2 Elimination behavior

Our proposed mechanism of sequential elimination relies on eliminations being guided by underlying preferences, yet participants might instead decide at the outset and eliminate options perfunctorily thereafter. To address this concern, we analyze elimination order in relation to option characteristics and the presentation order, focusing on participants' final submitted sequences to facilitate comparison with examination sequences under the Direct Procedure.

We distinguish three option types based on monetary features: FSD-dominated options, inferior to another option within the same menu; boundary options, riskier and with the highest or lowest expected value and a zero payoff in one state (i.e., at the endpoints of the budget lines); and middle options, safer and with intermediate expected value and the smallest payoff spread (i.e., equal or nearly equal payoffs).²⁰ Figure 2 depicts distinct patterns in elimination sequences. FSD-dominated and boundary options are concentrated among early eliminations, whereas middle options persist longer and account for a larger share of final choices. Among low-IQ participants (left panel), 43.1% and 50.0% of first eliminations are FSD-dominated and boundary options, respectively,

²⁰FSD-dominated and boundary options overlap, as one boundary option may be FSD-dominated by the other in the same menu.

compared with only 3.6% for middle options; by the final choice, the share of middle options rises to 36.5%. The pattern is similar among high-IQ participants (middle panel) and in the overall sample (right panel).

A comparison with the Direct Procedure sharpens this pattern. We record position as the order in which an option is eliminated under Sequential Elimination or examined under the Direct Procedure, with lower values indicating earlier elimination or examination. For the Direct Procedure, the mean position of each option type clusters around the midpoint (6).²¹ Under Sequential Elimination, FSD-dominated and boundary options are eliminated before the midpoint (4.807 and 4.129), whereas middle options survive beyond it (8.103). These between-procedure differences are broadly similar across IQ groups. Regarding the role of presentation order, the Spearman correlation between elimination or examination order and presentation order is 0.730 under the Direct Procedure but only 0.085 under Sequential Elimination (Appendix Table D.14, Column 4).

Regression estimates confirm these option-characteristic patterns and the presentation-order contrast (Appendix Table D.15). These results suggest a deliberate, preference-guided elimination process that accords with our framework.

5.3 Final choice patterns

Having established the elimination patterns, we examine whether they persist in final choices. Regression estimates in Columns 1–3 of Table 4 show a marginally significant reduction in the probability that low-IQ participants choose FSD-dominated options under Sequential Elimination, and a larger, statistically significant reduction under the Minimum-Time Procedure (see also Appendix Table D.16 for descriptive statistics). Neither treatment significantly changes their probability of choosing boundary or middle options. In other words, the treatment effects appear concentrated in reducing choices of FSD-dominated options rather than in broader shifts across option risk profiles. Under the Direct Procedure, high-IQ participants are less likely than low-IQ participants to choose FSD-dominated options, consistent with their greater observed rationality. Column 4

²¹Because choice sets contain multiple FSD-dominated and boundary options, we calculate their mean position for each participant–problem observation; the middle option’s position is taken directly, as it is unique within each choice set. See Columns 1–3 of Appendix Table D.14.

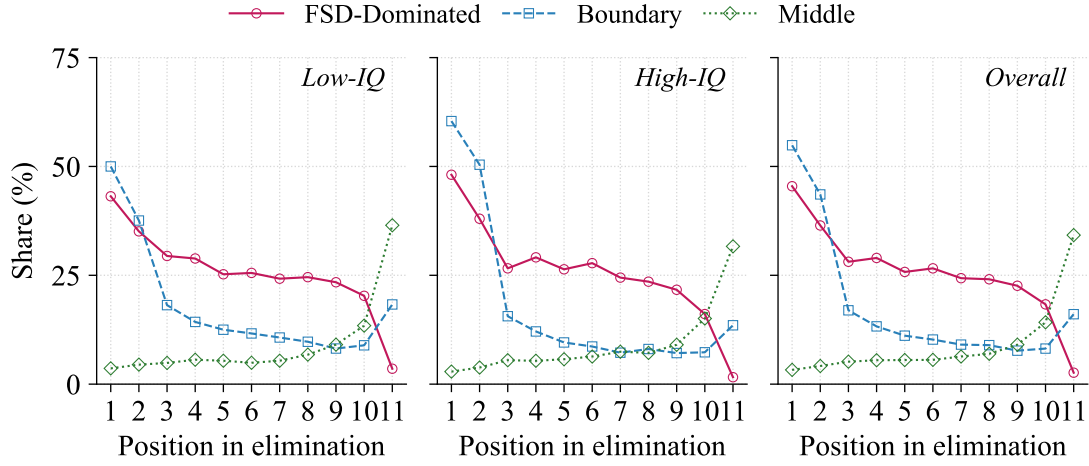


FIGURE 2. Elimination behavior across positions under Sequential Elimination. Each panel reports the shares of FSD-dominated, boundary, and middle options at each position in the final elimination sequences for low-IQ participants (left panel), high-IQ participants (middle panel), and the overall sample (right panel). The final position (11) corresponds to the chosen option.

finds no significant treatment difference in the presentation position of chosen options, weighing against a limited-search account in which rationality deviations arise from a tendency to choose earlier-listed options.

In addition, we exploit a symmetry in our design: eleven pairs of decision problems have the same lottery menus but reversed payoff dimensions; corresponding pairs are referred to as “mirrored” pairs (e.g., $[168, 0]$ and $[0, 168]$ in problems 1 and 10, respectively). The frequency of choosing such mirrored options signals consistency across these paired presentations, although deviations need not violate GARP and may reflect stochastic choice. Among low-IQ participants, mirrored-choice frequency rises from 44.0% under the Direct Procedure to 51.7% under Sequential Elimination, with a marginally significant estimate in Column 5, whereas no such shift is observed among high-IQ participants. These results provide suggestive evidence that Sequential Elimination may reduce choice stochasticity, an effect beyond the scope of our framework but complementary to our main results on economic rationality.

TABLE 4. Effects of choice procedures on final choice patterns.

	FSD-dominated (1)	Boundary (2)	Middle (3)	Presentation position (4)	Mirrored options chosen (5)
Sequential Elimination	-0.476 (0.255)	0.488 (0.310)	-0.011 (0.198)	-0.017 (0.015)	0.315 (0.161)
Minimum-Time Procedure	-0.880 (0.278)	-0.019 (0.307)	-0.247 (0.202)	-0.017 (0.016)	0.022 (0.152)
High-IQ	-1.059 (0.364)	0.033 (0.330)	0.047 (0.198)	-0.029 (0.018)	0.250 (0.155)
Sequential Elimination $\times$ high-IQ	0.285 (0.478)	-0.391 (0.458)	-0.213 (0.279)	0.031 (0.024)	-0.372 (0.220)
Minimum-Time Procedure $\times$ high-IQ	0.723 (0.457)	0.526 (0.439)	-0.187 (0.287)	0.025 (0.024)	-0.021 (0.217)
Controls	Yes	Yes	Yes	Yes	Yes
Decision problem fixed effects	Yes	Yes	Yes	Yes	Yes
Log α				-1.923 (0.025)	
Observations	14784	14784	14784	14784	6776

Note: This table reports the regression results on the effects of Sequential Elimination and the Minimum-Time Procedure on final choice patterns, relative to the Direct Procedure. Columns 1–3 present logit coefficients, with binary indicators for the corresponding choice features as dependent variables. Column 4 presents negative binomial estimates, with the presentation position of the chosen option as the dependent variable. Column 5 presents logit coefficients estimated on the subsample of eleven symmetrically equivalent pairs of decision problems, in which each option has a mirrored counterpart with reversed dimensions; the dependent variable indicates whether the participant chose the mirrored option in both problems of a pair. Observations are at the participant–problem level in Columns 1–4 and the participant–pair level in Column 5. Controls include ASRS inattention, selective attention, working memory, attitude toward inconsistency, age, gender, education, and decision order. All specifications include a constant. Standard errors clustered at the individual level are reported in parentheses.

5.4 Procedural preferences

Our pilot evidence offers suggestive insights into preferences over procedures and their relationship with economic rationality. In an additional treatment, participants choose between the Direct Procedure and Sequential Elimination after practicing both. Low-IQ participants are more likely to choose Sequential Elimination, with 82.1% doing so compared with 47.2% of high-IQ participants; IQ is negatively associated with choosing it (Appendix Table D.17). Differences in economic rationality between participants who choose and those assigned to the same procedure are statistically nonsignificant (Appendix Table D.18, Panel A). Likewise, counterfactual predictions for low-IQ participants under assigned procedures show no evident change in predicted rationality when evaluated as if they had chosen the same procedures voluntarily (Panel B).

Despite being correlational and based on a limited design, the preference for Sequential Elimination and the associated rationality improvements among low-IQ participants point to a potential alignment between the descriptive and normative roles of choice procedures. This is consistent with the possibility that participants appreciate its rationality–time trade-off or view it as a commitment device for more deliberate choice, such that merely offering it may suffice. Whether individuals internalize their cognitive constraints and evaluate the trade-offs between available choice procedures remains an open question with implications for policy design.

6 Related literature

This paper builds on models of limited attention (e.g., [Masatlioglu, Nakajima, and Ozbay, 2012](#)), which motivate our direct-choice benchmark. A related strand studies the “attention overload” restriction, whereby alternatives receive weakly less attention as the choice set expands ([Lleras et al., 2017](#); [Cattaneo et al., 2026](#)), while [Cattaneo et al. \(2020\)](#) allow an alternative to receive less attention when the choice set contracts. Whereas this literature primarily characterizes choice under such constraints, we ask whether alternative choice procedures can improve economic rationality despite limited attention, while accounting for cognitive heterogeneity linked by [Dardanoni et al. \(2020\)](#) to variation in consideration-set size. More specifically, we introduce the minimum attention property, which characterizes a lower bound on the size of consideration sets, thereby ensuring that eliminations remain grounded in preference-maximizing comparisons.

The paper also relates to elimination-based choice models. Early contributions model elimination based on attributes ([Tversky, 1972](#)) or environmental cues ([Gigerenzer, Todd, and ABC Research Group, 2000](#)). More recently, [Masatlioglu and Nakajima \(2007\)](#) describe a DM with potentially incomplete preferences who applies menu-dependent dominance criteria and selects all surviving alternatives, while our sequential elimination operates through menu-independent criteria until a single option remains. Other models characterize elimination through multiple acyclic relations ([Manzini and Mariotti, 2007](#)), checklists of desirable properties ([Mandler, Manzini, and Mariotti, 2012](#)), or ordered binary comparisons ([Apesteguia and Ballester, 2013](#)). These approaches accommodate

choice inconsistencies through non-standard preferences or model-specific elimination rules. Absent an explicit link between procedures and cognitive constraints, however, they do not generally sign the effect of sequential elimination on choice consistency. Our framework instead maintains standard preferences and shows how procedural design can shape choice consistency through sequential elimination.

An extensive literature in marketing and psychology documents systematic differences between choice and rejection procedures (e.g., [Shafir, 1993](#); [Dhar and Wertenbroch, 2000](#)). Elimination can broaden consideration ([Yaniv and Schul, 1997](#)), induce distinct decision criteria ([Yaniv and Schul, 2000](#)), promote deliberative processing ([Sokolova and Krishna, 2016](#)), and enhance decision confidence ([Chan and Wang, 2018](#)). In economics, [Besedeš et al. \(2015\)](#) show that a *sequential tournament*—in which participants choose from four groups of four options before selecting among the preliminary winners—increases optimal choice relative to direct choice from all sixteen options. Our framework can accommodate this result by interpreting the tournament as repeated applications of the direct-choice benchmark to these smaller menus, provided that the DM can consider at least four options but not all sixteen. Sequential Elimination, in contrast, does not require prespecified grouping and ordering of alternatives, which may lower implementation costs and mitigate context effects.

Finally, we contribute to research on the determinants of economic rationality. A broader literature documents associations with cognitive ability, market experience, and socioeconomic characteristics ([List and Millimet, 2008](#); [Burks et al., 2009](#); [Choi et al., 2014](#); [Dean and Martin, 2016](#); [Cappelen et al., 2023](#); [Echenique, Imai, and Saito, 2023](#)). Evidence on interventions remains limited and mixed: [Kim et al. \(2018\)](#) find that education improves rationality by enhancing cognitive ability, whereas [Banks, Carvalho, and Perez-Arce \(2019\)](#) find no significant effect of an extension of compulsory schooling. [Halevy and Mayraz \(2024\)](#) document a preference for rule-based over case-by-case investment decisions, particularly when the rules are simple. We complement this literature with causal evidence that a simple choice procedure alleviates the impact of cognitive constraints on economic rationality.

7 Concluding remarks

This paper demonstrates that sequential elimination, an intuitive choice procedure grounded in economics and the cognitive sciences, can improve economic rationality. Guided by a tractable framework, we obtain causal evidence for a sequential elimination effect among participants with lower cognitive ability and among those facing greater attentional constraints in a randomized controlled experiment involving risky decision-making. The evidence is consistent with a mechanism in which the procedure mitigates the impact of limited attention through deliberate, preference-guided eliminations.

Sequential elimination entails relatively modest implementation costs, as it involves neither a calibrated time floor nor prespecified menu arrangements and can in principle be applied to any finite choice set, covering a broad range of consequential decisions. Our framework clarifies why individuals need only manage binary comparisons under well-behaved preferences for the procedure to be effective. Our empirical evidence further suggests that ASRS may offer a low-cost proxy for identifying individuals who could benefit from sequential elimination. Together, these features underscore the potential of sequential elimination as a scalable intervention.

Field studies of sequential elimination present a promising avenue, particularly in high-stakes settings such as retirement planning and healthcare decisions. Beyond risky choice, examining the robustness of the procedure across domains such as consumer goods, intertemporal choices, and altruistic choices may help assess its generalizability. Future implementations could explore adaptive designs that enhance efficiency by tailoring the procedure to individual needs, for example, by allowing individuals with greater capacity to transition to direct choice earlier. Such efforts may catalyze the development of new choice procedures with broader applicability across diverse populations.

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Supplemental appendix

A Theoretical details

A.1 Proof of Proposition 1

Let c be a choice function. Consider the following three conditions:

- [1] c satisfies GARP.
- [2] There exists a preference relation $\succeq$ on X such that for all $A \in \mathcal{X}$, $c(A) \in \{x \in A \mid x \succeq y, \forall y \in A\}$.
- [3] c is a choice by sequential elimination.

By Theorem 1 of [Nishimura, Ok, and Quah \(2017\)](#), [1] holds if and only if [2] holds. It therefore suffices to show that [3] holds if and only if [2] holds.

[3] implies [2]. Suppose that [3] holds. Let $\succeq$, γ , and e be the preference relation, consideration set mapping satisfying the minimum attention property, and elimination function satisfying the conditions in Definition 3. Suppose, for contradiction, that there exists some A such that $c(A) = e_{|A|}(A) \notin \{x \in A \mid x \succeq y, \forall y \in A\}$. Since $\succeq$ is complete and transitive, $\{x \in A \mid x \succeq y, \forall y \in A\} \neq \emptyset$, and all $\succeq$ -best options must be eliminated before the final round. Let $r \in \{1, \dots, |A| - 1\}$ be the last round at which a $\succeq$ -best option is eliminated. Then $\{x \in A_r \mid x \succeq e_r(A), x \neq e_r(A)\} = \emptyset$, and hence $\{x \in \gamma(A_r) \mid x \succeq e_r(A), x \neq e_r(A)\} = \emptyset$, contradicting Definition 3(ii). Therefore, for all $A \in \mathcal{X}$, $c(A) \in \{x \in A \mid x \succeq y, \forall y \in A\}$.

[2] implies [3]. Suppose that [2] holds. Define $\gamma(A) = A$ for all $A \in \mathcal{X}$, which clearly satisfies the minimum attention property. Given c , construct an elimination function e such that for all $A \in \mathcal{X}$: if $|A| \geq 2$, then $e(A) = (e_1(A), \dots, e_{|A|-1}(A), c(A))$ with $\bigcup_{r=1}^{|A|} \{e_r(A)\} = A$; if $|A| = 1$, then $e_1(A) = c(A)$. For all A and $r = 1, \dots, |A|$, we have $e_r(A) \in \gamma(A_r)$ (Definition 3(i)); $\{x \in \gamma(A_r) \mid x \succeq e_r(A), x \neq e_r(A)\} \neq \emptyset$ if $|A_r| \geq 2$ (Definition 3(ii)); and $e_{|A|}(A) = c(A)$ (Definition 3(iii)). Thus, c is a choice by sequential elimination.

A.2 Consistency in compliance with FSD

Because the choice options in our experiment involve only two states with equal probability, we focus on $X \subseteq \mathbb{R}_+^2$ and let $\mathcal{X}$ be the set of all nonempty subsets of X . For any two options $x, y \in X$ defined by $x = (x_1, x_2)$ and $y = (y_1, y_2)$, we write $x \geq_F y$ if $(x_1, x_2) \geq (y_1, y_2)$ or $(x_2, x_1) \geq (y_1, y_2)$; and $x >_F y$ if $x \geq_F y$ but not $y \geq_F x$.

A preference relation $\succeq$ is said to be *first-order stochastically monotone* if $x \geq_F y$ implies $x \succeq y$ and $x >_F y$ implies $x \succeq y$ but not $y \succeq x$. For any x, y , we write $x R_F^D y$ ($x R_F^S y$, respectively) if there exist menus $A, B \in \mathcal{X}$ and an alternative $z \in A$ such that $c(A) = x$, $c(B) = y$, and $z \geq_F y$ ($z >_F y$, respectively). Let R_F denote the transitive closure of R_F^D . We define FSD-GARP formally as follows.

Definition 4 (FSD-GARP). A choice function c is said to satisfy FSD-GARP if, for any $x, y \in X$, $x R_F y$ implies that $y R_F^S x$ does not hold.

Any first-order stochastically monotone preference relation $\succeq$ on X and choice function c on $\mathcal{X}$ fall within the primitives of [Nishimura, Ok, and Quah \(2017\)](#). By their Theorem 1, c satisfies FSD-GARP if and only if there exists a first-order stochastically monotone preference relation $\succeq$ on X such that for all $A \in \mathcal{X}$, $c(A) \in \{x \in A \mid x \succeq y, \forall y \in A\}$. An ordered pair (x, y) constitutes an FSD-GARP violation if $x R_F y$ and $y R_F^S x$. By the same logic as in the proof of Proposition 1, c satisfies FSD-GARP if and only if it is a choice by sequential elimination with a first-order stochastically monotone preference relation, or equivalently, if it is consistent with maximizing such a preference relation.

B Experimental design details

B.1 Experimental Instructions

B.1.1 Introduction

Welcome to our study on decision-making.

The study consists of three parts. In Part 1, you will make a series of economic decisions. In Part 2, you will complete tasks measuring cognitive skills. In Part 3, you will respond to a short questionnaire about attention and everyday behaviour. Detailed instructions will be provided at the beginning of each part.

You will receive a £1 participation fee upon completing the study. Additionally, you may earn up to £14.60, depending partly on your decisions and partly on chance. You will be paid within five working days after completing the study.

During the study, we will use experimental tokens instead of pounds. The total number of tokens you earn will be converted into pounds at the following rate:

$$25 \text{ tokens} = £1$$

Please read the instructions carefully. To ensure your understanding and the validity of the data collected, you will be asked comprehension questions at the start. You will have two chances to answer each question correctly. If you are unsuccessful on both attempts, you will not be able to proceed with the study.

During the study, you may also encounter questions or tasks designed to verify instruction-following and data quality. It is important that you respond to these carefully. If you fail multiple quality checks throughout the study, your submission will not be eligible for compensation.

Important: Once you have moved on to the next question, you **cannot** go back and change your choice. Do not close the web browser at any time!

B.1.2 Part 1

In this part of the study, you will make 25 economic decisions, all following the same format. In each decision problem, you will be asked to choose **one** option out of

multiple options. An option [X, Y] indicates that you will receive either X tokens or Y tokens, with **equal** probability. For example, the option [24, 32] indicates that you will receive either 24 tokens or 32 tokens, each with a 50% probability.

You will earn real money based on your decisions. For each decision, carefully evaluate all available options and choose the one you prefer. There are no right or wrong answers. We are interested in studying your preferences.

At the end of the study, one of the 25 decisions will be randomly drawn by the computer, with each decision equally likely to be selected. You will earn tokens based on your choice in this drawn decision.

Please answer the following questions to confirm that you have understood the instructions.

1. For an option [X, Y], which one of the following statements is correct?

I will earn both X tokens and Y tokens at the same time; I will earn either X tokens or Y tokens with the same probability; I will be more likely to earn X tokens than to earn Y tokens.

2. Which one of the following statements is **not** correct?

I will select the decision problem for payment myself; Only one decision problem will be drawn for payment; The decision problem for payment will be randomly drawn by the computer.

When you are ready, click on 'Next' to proceed.

B.1.2.1 Direct Procedure

You will make decisions by a procedure called **sequential examination**. You will be asked to **examine** each option one by one by clicking on it. After examining all options, you will be asked to **choose** the single option that you **prefer** by clicking on it.

Below, you can see an example of a decision problem. A list of options appears on the left side of the screen. A **Choice List** on the right side of the screen displays the options you examined.

Example

Examine all the options.

Options

[48, 54]
[88, 24]
[72, 36]
[16, 78]

Choice List

For example, in the screen below, to mark the option [16, 78] as examined, you click on it, which moves it to the Choice List.

Example

Examine all the options.

Options

[48, 54]
[88, 24]
[72, 36]

Choice List

[16, 78]

Examine all available options. Once all are in the Choice List, choose your preferred option. Your choice will be highlighted in yellow. For example, in the screen below, suppose you chose [88, 24]. You can click on ‘Next’ to confirm this choice and proceed to the next problem.

Example

Choose the option that you **prefer** from the Choice List.

Options

Choice List

[48, 54]
[88, 24]
[72, 36]
[16, 78]

Next

Regarding payment, suppose you choose [88, 24]. If you are paid according to this choice, you will receive either 88 tokens or 24 tokens, each with a 50% probability.

To familiarise yourself with the procedure, you will first complete one practice problem. This will not affect your payment. After that, you will be asked to complete all the problems.

Please answer the following question to confirm your understanding of the instructions.

How is your choice determined in each decision problem?

By the option that appears at the top of the Choice List; By the option that appears at the bottom of the Choice List; By clicking on my preferred option in the Choice List.

When you are ready, click on 'Next' to proceed.

B.1.2.2 Sequential Elimination

You will make decisions by a procedure called **sequential elimination**. You will be asked to **eliminate** options one by one by clicking on those you **do not prefer**, until only one option remains. The **last remaining option** is your choice in the decision problem.

Below, you can see an example of a decision problem. A list of options appears on the left side of the screen. A **Trash** box on the right side of the screen displays the options you eliminated.

Example

Eliminate the options that you **do not prefer** from this list.

Options	Trash
[48, 54]	
[88, 24]	
[72, 36]	
[16, 78]	

For example, in the screen below, to eliminate the option [16, 78], you click on it, which moves it to the Trash. Note that you can recover options from the Trash by clicking on them. For instance, clicking on [16, 78] in the Trash will move it back to the Options

list.

Example
Eliminate the options that you **do not prefer** from this list.

Options
[48, 54]
[88, 24]
[72, 36]

Trash
[16, 78]

Eliminate options until only one option remains. Your choice will be highlighted in yellow. For example, in the screen below, suppose you eliminated [16, 78], [72, 36], and [48, 54]. As a result, the last remaining option, [88, 24], becomes your choice. You can click on 'Next' to confirm this choice and proceed to the next problem.

Example
Your choice is

Options
[88, 24]

Trash
[16, 78]
[72, 36]
[48, 54]

Next

Regarding payment, suppose you choose [88, 24]. If you are paid according to this choice, you will receive either 88 tokens or 24 tokens, each with a 50% probability.

To familiarise yourself with the procedure, you will first complete one practice problem. This will not affect your payment. After that, you will be asked to complete all the problems.

Please answer the following question to confirm your understanding of the instructions.

How is your choice determined in each decision problem?

By clicking on my preferred option in the Trash; By clicking on my preferred option

in the Options list; By eliminating the options I do not prefer to the Trash, until only one remains in the Options list.

When you are ready, click on 'Next' to proceed.

B.1.2.3 Minimum-Time Procedure

You will make decisions by a procedure where you must spend at least **35 seconds** on each decision problem. You will be asked to choose the single option that you prefer by clicking on it. The 'Next' button, which you use to submit your choice, will become available only after 35 seconds have elapsed since each problem appeared.

Below, you can see an example of a decision problem. A list of options appears on the left side of the screen. A **timer** on the right side of the screen displays the time elapsed since the problem appeared. In this example, only 5 seconds have elapsed, so 'Next' is not yet available.

Example

Choose the option that you **prefer** from this list.

Options

- [48, 54]
- [88, 24]
- [72, 36]
- [16, 78]

0005

Choose your preferred option from the list. Your choice will be highlighted in yellow. For example, in the screen below, suppose you chose [88, 24] and 35 seconds have elapsed. You can click on 'Next' to confirm this choice and proceed to the next problem.

Example

Choose the option that you **prefer** from this list.

Options
[48, 54]
[88, 24]
[72, 36]
[16, 78]

0035

Next

Regarding payment, suppose you choose [88, 24]. If you are paid according to this choice, you will receive either 88 tokens or 24 tokens, each with a 50% probability.

To familiarise yourself with the procedure, you will first complete one practice problem. This will not affect your payment. After that, you will be asked to complete all the problems.

Please answer the following question to confirm your understanding of the instructions.

How are you allowed to submit your choice in each decision problem?

By submitting immediately after the problem appears; By waiting 5 seconds after the problem appears; By waiting 35 seconds after the problem appears.

When you are ready, click on 'Next' to proceed.

1) Examine all the options.

Options	Choice List
68, 50	
100, 34	
152, 8	
0, 84	
34, 67	
118, 25	
134, 17	
84, 42	
16, 76	
168, 0	
56, 56	

(a) Initial screen

1) Examine all the options.

Options	Choice List
68, 50	
100, 34	
152, 8	
0, 84	0, 84
34, 67	
118, 25	
134, 17	
84, 42	
16, 76	
168, 0	
56, 56	

(b) An option is examined

1) Choose the option that you **prefer** from the Choice List.

Options	Choice List
	68, 50
	100, 34
	152, 8
	0, 84
	34, 67
	118, 25
	134, 17
	84, 42
	16, 76
	168, 0
	56, 56

(c) All options are examined

1) Choose the option that you **prefer** from the Choice List.

Options	Choice List
	68, 50
	100, 34
	152, 8
	0, 84
	34, 67
	118, 25
	134, 17
	84, 42
	16, 76
	168, 0
	56, 56

Next

(d) A choice is made

FIGURE B.1. Direct Procedure interface.

1) Eliminate the options that you **do not prefer** from this list.

Options	Trash
56, 56	
118, 25	
0, 84	
16, 76	
34, 67	
168, 0	
68, 50	
134, 17	
152, 8	
84, 42	
100, 34	

(a) Initial screen

1) Eliminate the options that you **do not prefer** from this list.

Options	Trash
56, 56	
118, 25	0, 84
16, 76	
34, 67	
168, 0	
68, 50	
134, 17	
152, 8	
84, 42	
100, 34	

(b) An option is eliminated

1) Your **choice** is

Options	Trash
	0, 84
	56, 56
	118, 25
	16, 76
	34, 67
	168, 0
	134, 17
	68, 50
	84, 42
	100, 34

152, 8

Next

(c) A choice is made

FIGURE B.2. Sequential Elimination interface.

1) Choose the option that you **prefer** from this list.

Options

- 34, 67
- 134, 17
- 168, 0
- 68, 50
- 118, 25
- 16, 76
- 152, 8
- 100, 34
- 84, 42
- 56, 56
- 0, 84

0005

(a) Initial screen

1) Choose the option that you **prefer** from this list.

Options

- 34, 67
- 134, 17
- 168, 0
- 68, 50
- 118, 25
- 16, 76
- 152, 8
- 100, 34
- 84, 42
- 56, 56
- 0, 84

0020

(b) A choice is made before 35 seconds

1) Choose the option that you **prefer** from this list.

Options

- 34, 67
- 134, 17
- 168, 0
- 68, 50
- 118, 25
- 16, 76
- 152, 8
- 100, 34
- 84, 42
- 56, 56
- 0, 84

0035

Next

(c) A choice is made after 35 seconds

FIGURE B.3. Minimum-Time Procedure interface.

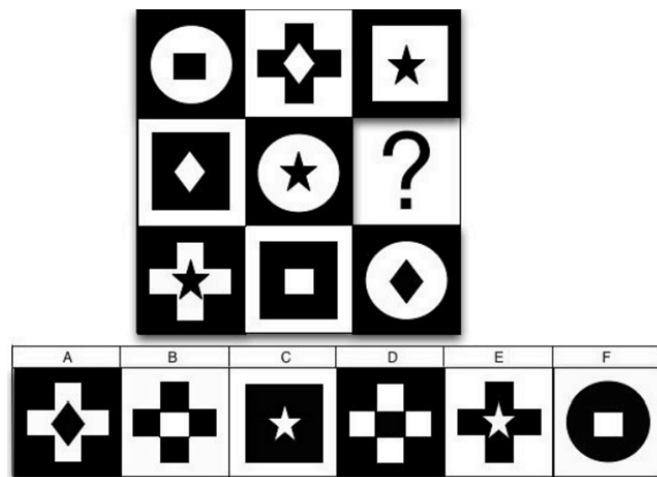
B.1.3 Part 2

This part consists of three tasks measuring cognitive skills. Your payment will depend on your performance in these tasks. Each task contains a different number of questions. At the end of the experiment, three questions will be randomly drawn from these tasks, each with an equal probability. You will earn 25 tokens for each correct answer among the drawn questions.

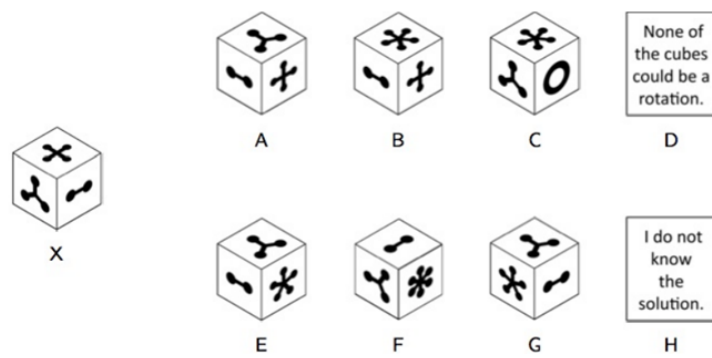
Task 1 (IQ). In this task, you are asked to answer 10 questions: five on matrix reasoning and five on three-dimensional rotation. Each question has one correct answer. You will have a maximum of 12 minutes to complete the task.

Task 2 (selective attention). This task measures your concentration. In each round, you are asked to identify the color of the word shown on the screen. The word itself is irrelevant—you can safely ignore it. To indicate the color of the word, please use the keys **r**, **g**, **b**, and **o** for **red**, **green**, **blue**, and **orange**, respectively. A plus sign will appear before each word; please keep your eyes on it. You will have only two seconds in each round. You will have two practice rounds, followed by 20 rounds to complete.

Task 3 (working memory). This task measures your working memory. In each round, you are asked to memorize a sequence of digits, ranging from four to eight digits. After the presentation, you are asked to indicate whether a certain digit was included in the sequence. Please press **y** if you think that the digit is in the sequence; otherwise, press **n**. If your answer is correct, a green circle will appear; otherwise, a red circle will appear. You are then asked to type in the entire sequence. You will have one practice round, followed by 10 rounds to complete.

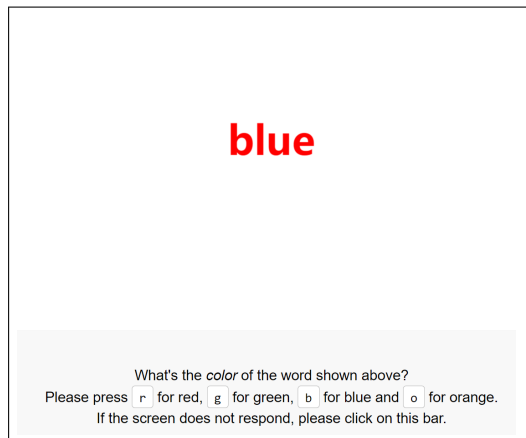


(a) Matrix reasoning problem



(b) Three-dimensional rotation problem

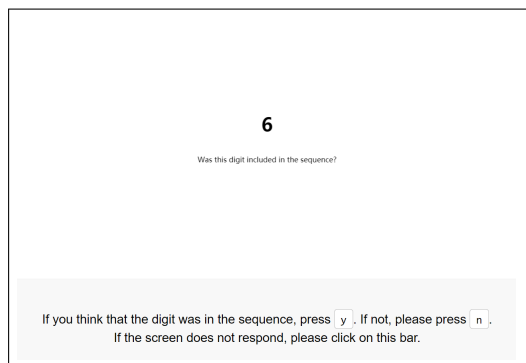
FIGURE B.4. Screenshots of Task 1.



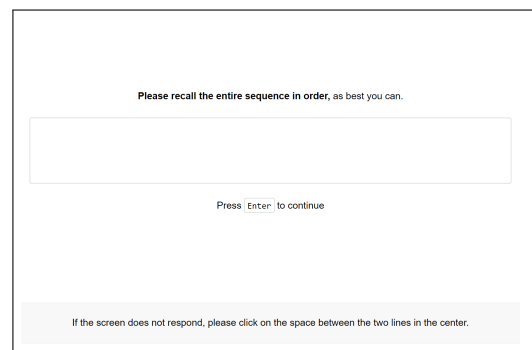
(a) Task 2



(b) Task 3 (memorization)



(c) Task 3 (recall 1)



(d) Task 3 (recall 2)

FIGURE B.5. Screenshots of Tasks 2 and 3.

B.1.4 Part 3

In this part, you will complete a self-report questionnaire consisting of three sets of questions about various scenarios. Please respond based on your own feelings in those situations. There are no right or wrong answers. We are interested in how you would feel.

Set A (attitude toward inconsistency). This question asks how you feel about a hypothetical choice scenario. Imagine that you are at a cinema and want to buy some popcorn. The cinema sells small tubs of popcorn for £3 and large ones for £7. In this scenario, you choose the small tub.

Now consider a different scenario. The cinema sells small tubs for £3, medium ones for £6.50 and large ones for £7. This time, you choose the large tub. In the first scenario, you prefer the small tub to the large one. In the second scenario, your choice suggests the opposite.

How at ease do you feel with your choices? Please rate this on the scale below. A rating of 0 means that you are not at all at ease with one or more of your choices and would very much like to make changes. A rating of 10 means that you are completely at ease and have no wish to change anything.

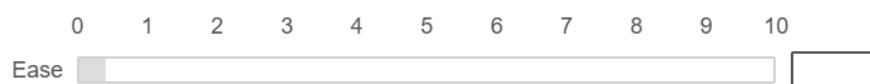


FIGURE B.6. Rating scale for attitudes toward inconsistency.

Set B (ASRS inattention subscale). This set of questions asks about your attention in everyday life. Please answer the questions below, rating yourself on each of the criteria shown. As you answer each question, select the response that best describes how you have felt and conducted yourself over the past 6 months.

TABLE B.1. ASRS inattention subscale.

No.	Item
1	How often do you make careless mistakes when you have to work on a boring or difficult project?
2	How often do you have difficulty keeping your attention when you are doing boring or repetitive work?
3	How often do you have difficulty concentrating on what people say to you, even when they are speaking to you directly?
4	How often do you have trouble wrapping up the final details of a project, once the challenging parts have been done?
5	How often do you have difficulty getting things in order when you have to do a task that requires organisation?
6	When you have a task that requires a lot of thought, how often do you avoid or delay getting started?
7	How often do you misplace or have difficulty finding things at home or at work?
8	How often are you distracted by activity or noise around you?
9	How often do you have problems remembering appointments or obligations?

Note: This table presents the 9-item inattention subscale from the ASRS version 1.1. Responses are recorded on a 5-point Likert scale: *never*, *rarely*, *sometimes*, *often*, and *very often*. Following the official guidelines, responses of *sometimes*, *often*, or *very often* are coded as one point for items 3, 4, 5, and 9. For the remaining items, only responses of *often* or *very often* are coded as one point.

Set C (ASRS hyperactivity–impulsivity subscale). This set of questions asks about your attention in everyday life. Please answer the questions below, rating yourself on each of the criteria shown. As you answer each question, select the response that best describes how you have felt and conducted yourself over the past 6 months.

TABLE B.2. ASRS hyperactivity-impulsivity subscale.

No.	Item
1	How often do you fidget or squirm with your hands or feet when you have to sit down for a long time?
2	How often do you leave your seat in meetings or other situations in which you are expected to remain seated?
3	How often do you feel restless or fidgety?
4	How often do you have difficulty unwinding and relaxing when you have time to yourself?
5	How often do you feel overly active and compelled to do things, like you were driven by a motor?
6	Please choose Very Often.
7	How often do you find yourself talking too much when you are in social situations?
8	When you're in a conversation, how often do you find yourself finishing the sentences of the people you are talking to, before they can finish them themselves?
9	How often do you have difficulty waiting your turn in situations when turn taking is required?
10	How often do you interrupt others when they are busy?

Note: This table presents the 9-item hyperactivity–impulsivity subscale from the ASRS version 1.1, with an additional instruction-following check (item 6). Responses are recorded on a 5-point Likert scale: *never*, *rarely*, *sometimes*, *often*, and *very often*. Item 6 is failed unless the response is *very often*. Following the official guidelines, responses of *sometimes*, *often*, or *very often* are coded as one point for items 2, 8, and 10. For the remaining subscale items, only responses of *often* or *very often* are coded as one point.

Demographics.

What is your age?

[Text-entry box]

What is your gender identity?

Male; Female; Other.

What is the highest level of education you have completed?

No formal qualifications; Secondary education (e.g., GED/GCSE); High school diploma/A-levels; Technical/community college; Undergraduate degree (BA/BSc/other); Graduate degree (MA/MSc/MPhil/other); Doctorate degree (PhD/other).

If you are currently studying, what level is it?

I am not currently studying; Secondary education (e.g., GED/GCSE); High school diploma/A-levels; Technical/community college; Undergraduate degree (BA/BSc/other); Graduate degree (MA/MSc/MPhil/other); Doctorate degree (PhD/other).

B.2 Decision problems

TABLE B.3. List of options in decision problems.

No.	Problem
1	[0, 84], [16, 76], [34, 67], [56, 56], [68, 50], [84, 42], [100, 34], [118, 25], [134, 17], [152, 8], [168, 0]
2	[0, 54], [20, 49], [44, 43], [68, 37], [88, 32], [108, 27], [132, 21], [152, 16], [172, 11], [196, 5], [216, 0]
3	[0, 225], [14, 204], [30, 180], [44, 159], [60, 135], [74, 114], [90, 90], [104, 69], [120, 45], [136, 21], [150, 0]
4	[0, 97], [18, 88], [36, 79], [50, 72], [64, 65], [92, 51], [112, 41], [134, 30], [154, 20], [176, 9], [194, 0]
5	[0, 108], [15, 96], [30, 84], [45, 72], [60, 60], [70, 52], [80, 44], [95, 32], [105, 24], [120, 12], [135, 0]
6	[0, 270], [6, 243], [12, 216], [18, 189], [24, 162], [30, 135], [36, 108], [42, 81], [48, 54], [54, 27], [60, 0]
7	[0, 150], [21, 136], [45, 120], [69, 104], [90, 90], [114, 74], [135, 60], [159, 44], [180, 30], [204, 14], [225, 0]
8	[0, 165], [17, 148], [33, 132], [50, 115], [66, 99], [83, 82], [100, 65], [116, 49], [133, 32], [149, 16], [165, 0]
9	[0, 102], [25, 92], [50, 82], [70, 74], [105, 60], [130, 50], [150, 42], [175, 32], [205, 20], [230, 10], [255, 0]
10	[0, 168], [8, 152], [17, 134], [25, 118], [34, 100], [42, 84], [50, 68], [56, 56], [67, 34], [76, 16], [84, 0]
11	[0, 216], [5, 196], [11, 172], [16, 152], [21, 132], [27, 108], [32, 88], [37, 68], [43, 44], [49, 20], [54, 0]
12	[0, 255], [10, 230], [20, 205], [32, 175], [42, 150], [50, 130], [60, 105], [74, 70], [82, 50], [92, 25], [102, 0]
13	[0, 90], [33, 79], [66, 68], [90, 60], [111, 53], [135, 45], [162, 36], [189, 27], [216, 18], [243, 9], [270, 0]

Note: Each option, denoted as $[x, y]$, yields either x or y tokens with equal probability.

TABLE B.3. List of options in decision problems (continued).

No.	Problem
14	[0, 270], [9, 243], [18, 216], [27, 189], [36, 162], [45, 135], [53, 111], [60, 90], [68, 66], [79, 33], [90, 0]
15	[0, 60], [27, 54], [54, 48], [81, 42], [108, 36], [135, 30], [162, 24], [189, 18], [216, 12], [243, 6], [270, 0]
16	[0, 194], [9, 176], [20, 154], [30, 134], [41, 112], [51, 92], [65, 64], [72, 50], [79, 36], [88, 18], [97, 0]
17	[0, 135], [12, 120], [24, 105], [32, 95], [44, 80], [52, 70], [60, 60], [72, 45], [84, 30], [96, 15], [108, 0]
18	[0, 58], [25, 53], [45, 49], [80, 42], [115, 35], [145, 29], [175, 23], [205, 17], [230, 12], [260, 6], [290, 0]
19	[0, 290], [6, 260], [12, 230], [17, 205], [23, 175], [29, 145], [35, 115], [42, 80], [49, 45], [53, 25], [58, 0]
20	[0, 195], [20, 175], [39, 156], [59, 136], [78, 117], [96, 99], [118, 77], [137, 58], [157, 38], [176, 19], [195, 0]
21	[0, 105], [14, 95], [28, 85], [42, 75], [56, 65], [77, 50], [91, 40], [105, 30], [119, 20], [133, 10], [147, 0]
22	[0, 120], [13, 110], [26, 100], [52, 80], [65, 70], [78, 60], [91, 50], [104, 40], [130, 20], [143, 10], [156, 0]
23	[0, 147], [10, 133], [20, 119], [30, 105], [40, 91], [50, 77], [65, 56], [75, 42], [85, 28], [95, 14], [105, 0]
24	[0, 156], [10, 143], [20, 130], [40, 104], [50, 91], [60, 78], [70, 65], [80, 52], [100, 26], [110, 13], [120, 0]
25	[0, 0], [11, 11], [22, 22], [33, 33], [44, 44], [55, 55], [66, 66], [77, 77], [88, 88], [99, 99], [111, 111]

Note: Each option, denoted as $[x, y]$, yields either x or y tokens with equal probability.

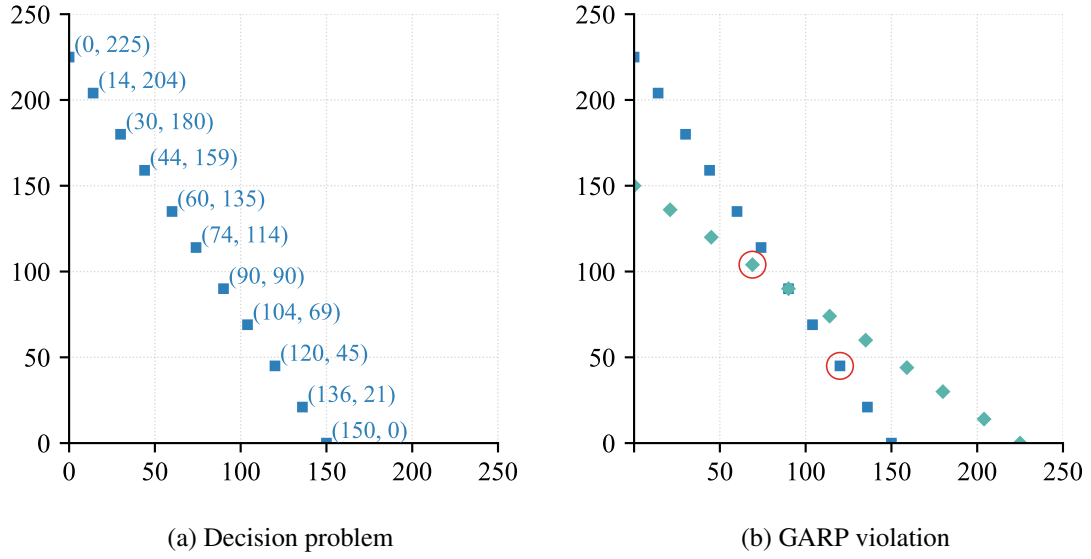


FIGURE B.7. Graphical illustrations of a decision problem and a GARP violation. In panel (a), squares represent a menu of options in two-dimensional space, randomly ordered during the experiment. In panel (b), squares and diamonds depict two distinct menus, with circles indicating a pair of choices that violate consistency with preference maximization, or equivalently, a GARP violation.

C Sample details

TABLE C.1. Sample descriptive statistics.

Variable	Direct Procedure	Sequential Elimination	Minimum-Time Procedure	Overall	
				Mean	<i>p</i> -value
Age	45.080 (0.912)	42.378 (0.880)	44.463 (0.942)	43.995 (0.528)	0.103
Female	0.533 (0.034)	0.552 (0.035)	0.483 (0.035)	0.523 (0.020)	0.352
Education	2.797 (0.079)	2.726 (0.075)	2.872 (0.085)	2.799 (0.046)	0.349
IQ	4.462 (0.146)	4.547 (0.146)	4.522 (0.156)	4.510 (0.086)	0.870
Selective attention	17.175 (0.232)	17.876 (0.219)	17.581 (0.230)	17.537 (0.132)	0.096
Working memory	6.425 (0.171)	6.537 (0.156)	6.345 (0.169)	6.435 (0.095)	0.769
ASRS inattention	2.448 (0.173)	2.756 (0.177)	2.754 (0.167)	2.649 (0.099)	0.190
ASRS hyperactivity-impulsivity	1.929 (0.149)	2.035 (0.139)	1.941 (0.155)	1.968 (0.085)	0.316
Attitude toward inconsistency	5.651 (0.208)	6.050 (0.194)	5.877 (0.205)	5.856 (0.117)	0.468
Observations	212	201	203	616	

Note: This table reports treatment-specific means of key variables and the corresponding overall mean. The last column presents *p*-values from Kruskal–Wallis tests of equality across procedures. The variable *education* is coded numerically based on the highest level of education attained: 0 = “no formal qualifications”, 1 = “high school diploma / A-levels/ secondary education”, 2 = “technical / community college”, 3 = “undergraduate degree”, 4 = “graduate degree”, 5 = “doctorate degree”. Selective attention is measured using a Stroop task; working memory using a Sternberg task; and inattention using the ASRS subscale. Standard errors are reported in parentheses.

TABLE C.2. Breakdown of observations by IQ and attention groups.

	Direct Procedure	Sequential Elimination	Minimum-Time Procedure	Overall
<i>Panel A: Overall by IQ</i>				
Low-IQ	118 (55.7%)	107 (53.2%)	107 (52.7%)	332 (53.9%)
High-IQ	94 (44.3%)	94 (46.8%)	96 (47.3%)	284 (46.1%)
Total	212 (100%)	201 (100%)	203 (100%)	616 (100%)
<i>Panel B: Overall by attention</i>				
Low-attention	108 (50.9%)	117 (58.2%)	127 (62.6%)	352 (57.1%)
High-attention	104 (49.1%)	84 (41.8%)	76 (37.4%)	264 (42.9%)
<i>Panel C: Low-attention participants by IQ</i>				
Low-IQ	60 (55.6%)	62 (53.0%)	67 (52.8%)	189 (53.7%)
High-IQ	48 (44.4%)	55 (47.0%)	60 (47.2%)	163 (46.3%)
Total	108 (100%)	117 (100%)	127 (100%)	352 (100%)
<i>Panel D: High-attention participants by IQ</i>				
Low-IQ	58 (55.8%)	45 (53.6%)	40 (52.6%)	143 (54.2%)
High-IQ	46 (44.2%)	39 (46.4%)	36 (47.4%)	121 (45.8%)
Total	104 (100%)	84 (100%)	76 (100%)	264 (100%)

Note: The table reports counts by IQ and attention groups for each treatment, with percentages in parentheses. Panels A and B show the overall breakdown by IQ and attention, respectively. Panel C reports counts and percentages for low-attention participants, while Panel D reports those for high-attention participants. Percentages are calculated within each panel.

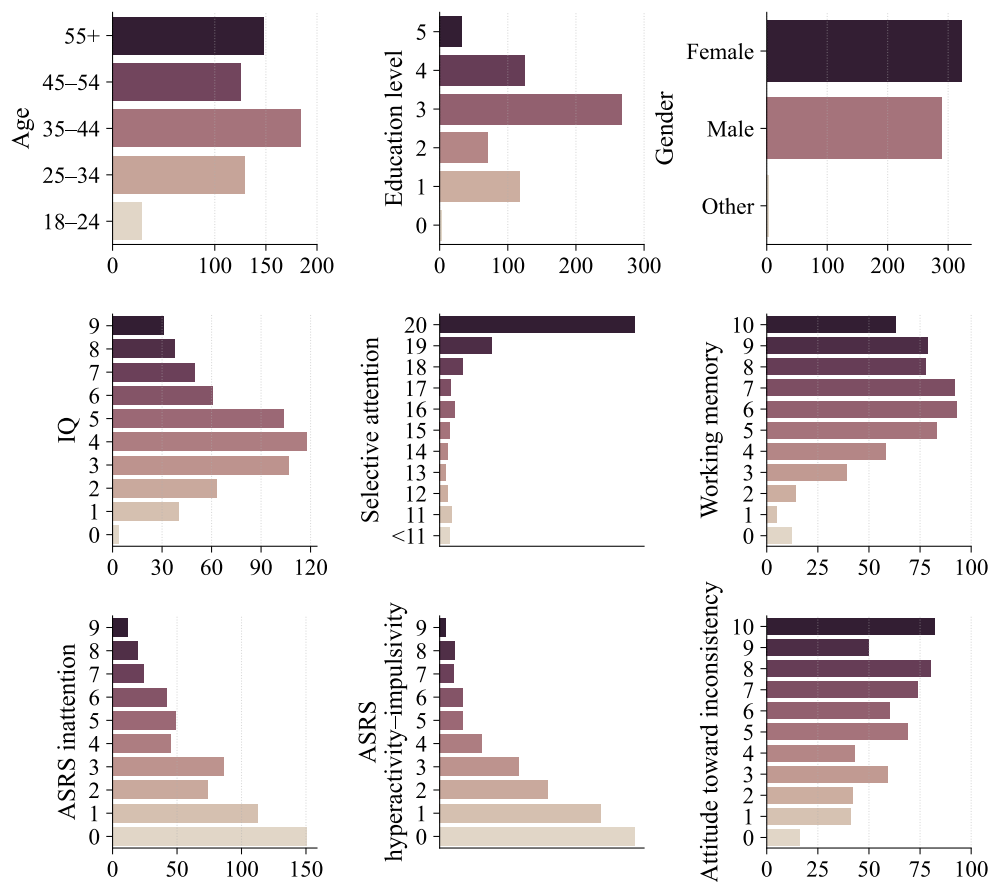


FIGURE C.1. Histograms of individual characteristics. The variable *education* is coded by the highest level attained: 0 = “no formal qualifications”, 1 = “high school diploma / A-levels / secondary education”, 2 = “technical / community college”, 3 = “undergraduate degree”, 4 = “graduate degree”, 5 = “doctorate degree”.

D Supplementary experimental results

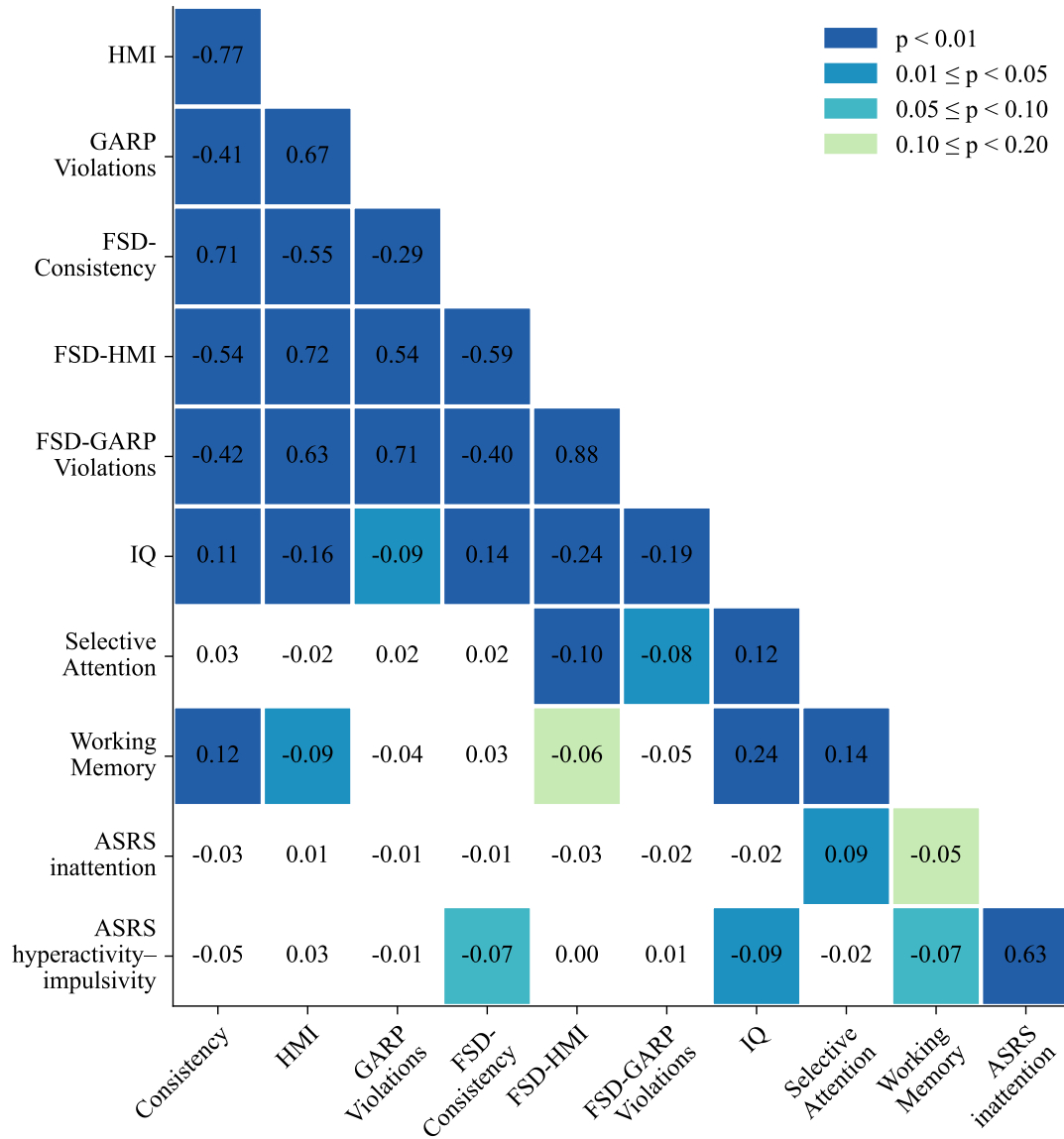


FIGURE D.1. Heatmap of correlations among economic rationality and cognitive measures. Pearson correlation coefficients appear in the lower triangular part of the matrix. Cells are color-coded by p-value, with darker colors indicating smaller p-values and white indicating $p \geq 0.20$.

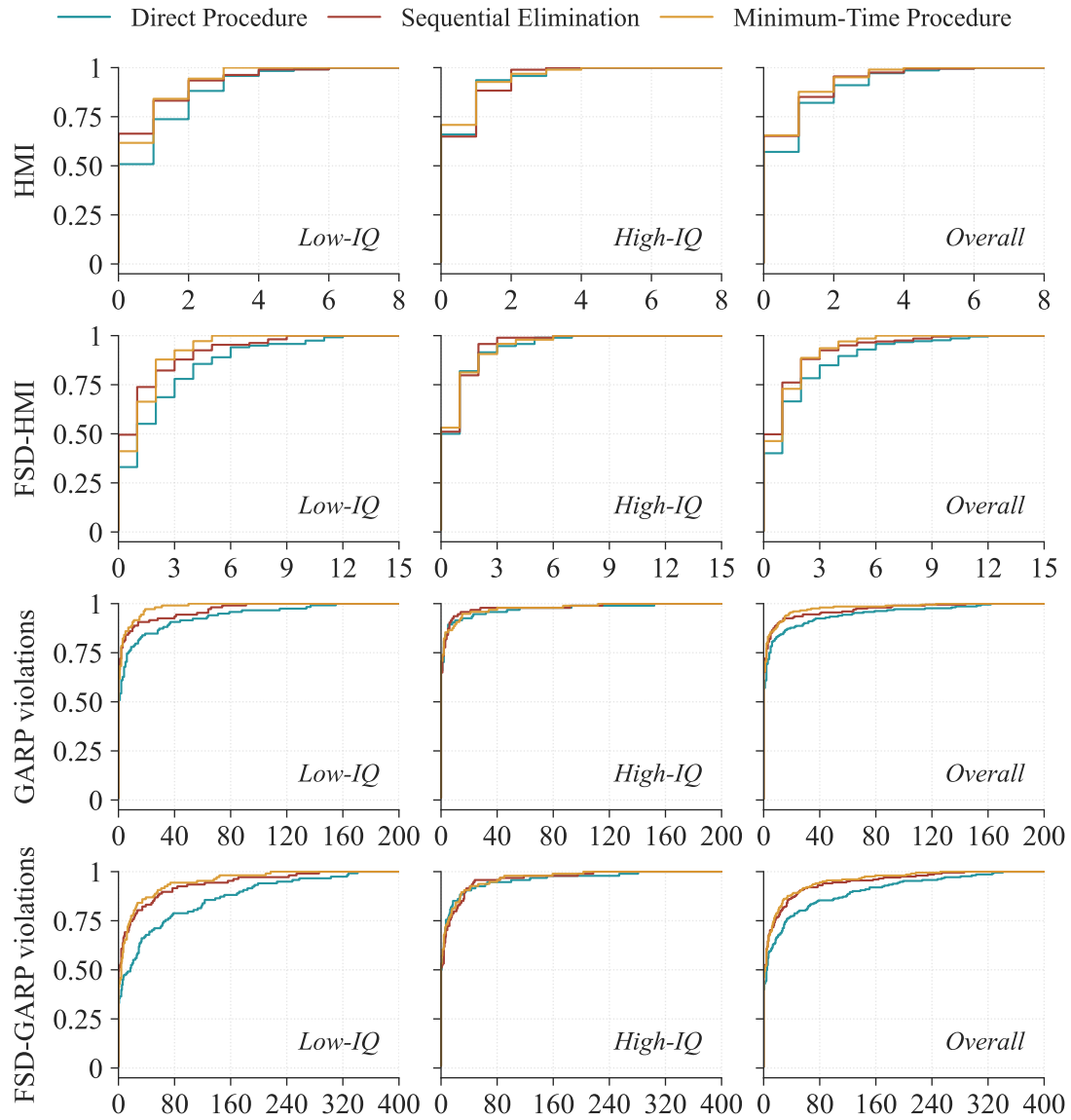


FIGURE D.2. Empirical cumulative distributions of rationality measures across treatments and IQ groups.

TABLE D.1. Pairwise treatment comparisons by IQ group across rationality measures.

Measure	SE vs. DP	SE vs. MTP	MTP vs. DP
<i>Panel A: Low-IQ participants</i>			
Consistency	0.154 (0.020)	0.047 (0.479)	0.107 (0.106)
HMI	-0.308 (0.021)	0.056 (0.642)	-0.364 (0.052)
GARP violations	-6.447 (0.011)	3.421 (0.731)	-9.867 (0.014)
FSD-consistency	0.164 (0.012)	0.084 (0.216)	0.080 (0.213)
FSD-HMI	-0.853 (0.003)	0.140 (0.377)	-0.993 (0.016)
FSD-GARP violations	-27.566 (0.001)	5.234 (0.523)	-32.800 (0.003)
<i>Panel B: High-IQ participants</i>			
Consistency	-0.011 (0.879)	-0.060 (0.383)	0.049 (0.472)
HMI	0.021 (0.743)	0.073 (0.356)	-0.052 (0.534)
GARP violations	-1.638 (0.950)	-0.342 (0.528)	-1.296 (0.544)
FSD-consistency	0.011 (0.884)	-0.021 (0.774)	0.031 (0.664)
FSD-HMI	-0.106 (0.915)	-0.023 (0.873)	-0.083 (0.764)
FSD-GARP violations	-2.681 (0.848)	1.167 (0.599)	-3.848 (0.722)
<i>Panel C: Overall</i>			
Consistency	0.081 (0.093)	-0.003 (0.942)	0.084 (0.079)
HMI	-0.166 (0.095)	0.065 (0.818)	-0.232 (0.053)
GARP violations	-4.363 (0.044)	1.652 (0.885)	-6.015 (0.026)
FSD-consistency	0.096 (0.049)	0.034 (0.489)	0.062 (0.202)
FSD-HMI	-0.535 (0.012)	0.066 (0.523)	-0.600 (0.048)
FSD-GARP violations	-16.800 (0.012)	3.369 (0.870)	-20.168 (0.015)

Note: The table presents the differences in means for pairwise treatment comparisons across rationality measures among low-IQ participants, high-IQ participants, and the overall sample. The corresponding p-values are reported in parentheses beside each difference. SE = Sequential Elimination, DP = Direct Procedure, MTP = Minimum-Time Procedure. Binary measures use chi-square tests; count measures use Mann–Whitney *U* tests.

TABLE D.2. Comparison of economic rationality between high-IQ and low-IQ participants.

	Direct Procedure	Sequential Elimination	Minimum-Time Procedure	Overall
Consistency	0.149 (0.030)	-0.016 (0.814)	0.090 (0.177)	0.078 (0.048)
HMI	-0.500 (0.005)	-0.170 (0.883)	-0.188 (0.123)	-0.294 (0.007)
GARP violations	-6.815 (0.007)	-2.006 (0.951)	1.757 (0.276)	-2.537 (0.020)
FSD-consistency	0.167 (0.013)	0.014 (0.843)	0.119 (0.089)	0.103 (0.010)
FSD-HMI	-1.267 (< 0.001)	-0.520 (0.377)	-0.357 (0.040)	-0.735 (< 0.001)
FSD-GARP violations	-35.914 (< 0.001)	-11.029 (0.540)	-6.963 (0.070)	-18.647 (< 0.001)
Observations (high-IQ/low-IQ)	94/118	94/107	96/107	284/332

Note: This table compares economic rationality between high-IQ and low-IQ participants within each treatment and in the overall sample. Mean differences (high-IQ minus low-IQ) are reported with p -values in parentheses. These p -values are derived from chi-square tests for choice consistency and FSD-consistency, and Mann–Whitney U tests for the remaining measures.

TABLE D.3. Regression results on the effects of choice procedures on economic rationality with treatment-by-high-IQ interactions.

	Consistency (1)	HMI (2)	GARP violations (3)	FSD-compliant		
				Consistency (4)	HMI (5)	GARP violations (6)
Sequential Elimination	0.605 (0.278)	-0.351 (0.213)	-0.689 (0.354)	0.696 (0.279)	-0.466 (0.187)	-0.759 (0.271)
Minimum-Time Procedure	0.459 (0.277)	-0.440 (0.187)	-1.285 (0.371)	0.345 (0.282)	-0.566 (0.167)	-0.904 (0.280)
High-IQ	0.428 (0.293)	-0.571 (0.221)	-0.676 (0.422)	0.639 (0.287)	-0.771 (0.198)	-1.071 (0.320)
Sequential Elimination $\times$ high-IQ	-0.514 (0.416)	0.298 (0.328)	0.405 (0.645)	-0.568 (0.405)	0.318 (0.292)	0.622 (0.467)
Minimum-Time Procedure $\times$ high-IQ	-0.218 (0.418)	0.332 (0.325)	1.232 (0.602)	-0.210 (0.405)	0.495 (0.278)	0.746 (0.475)
Selective attention	0.010 (0.028)	-0.007 (0.020)	0.008 (0.035)	0.014 (0.026)	-0.038 (0.018)	-0.037 (0.024)
Working memory	0.090 (0.038)	-0.043 (0.027)	-0.079 (0.051)	-0.001 (0.036)	-0.009 (0.023)	-0.035 (0.037)
ASRS inattention	-0.023 (0.035)	0.003 (0.024)	0.026 (0.048)	-0.006 (0.035)	-0.009 (0.023)	0.004 (0.036)
Attitude toward inconsistency	-0.030 (0.030)	0.007 (0.022)	-0.035 (0.039)	-0.021 (0.029)	0.018 (0.020)	0.012 (0.033)
Age	0.010 (0.007)	-0.008 (0.006)	-0.017 (0.010)	0.012 (0.007)	-0.009 (0.004)	-0.008 (0.007)
Female	-0.120 (0.172)	0.142 (0.132)	0.159 (0.248)	-0.087 (0.166)	-0.005 (0.115)	0.148 (0.189)
Education	0.113 (0.075)	-0.108 (0.056)	-0.159 (0.108)	0.061 (0.074)	-0.040 (0.049)	-0.030 (0.082)
Constant	-1.125 (0.754)	0.868 (0.513)	4.206 (0.972)	-1.485 (0.719)	1.909 (0.457)	5.151 (0.709)
Log α		0.003 (0.187)	2.145 (0.077)		0.056 (0.119)	1.748 (0.067)
Log-likelihood	-397.347	-665.999	-1244.996	-414.590	-946.968	-2022.552
Observations	616	616	616	616	616	616

Note: This table reports the regression coefficients for the effects of Sequential Elimination and the Minimum-Time Procedure on economic rationality relative to the Direct Procedure, estimated from models with treatment–High-IQ interactions. Columns 1 and 4 present coefficients from logistic regressions, while the remaining columns present those from negative binomial regressions. Robust standard errors are reported in parentheses.

TABLE D.4. Regression results on the effect of choice procedures on economic rationality with treatment-by-IQ interactions.

	Consistency (1)	HMI (2)	GARP violations (3)	FSD-compliant		
				Consistency (4)	HMI (5)	GARP violations (6)
Sequential Elimination	0.030 (0.484)	-0.098 (0.343)	-0.216 (0.645)	0.663 (0.496)	-0.352 (0.296)	-0.772 (0.434)
Minimum-Time Procedure	0.526 (0.474)	-0.760 (0.343)	-2.661 (0.619)	0.568 (0.481)	-0.977 (0.298)	-1.927 (0.465)
IQ	0.064 (0.068)	-0.129 (0.053)	-0.286 (0.104)	0.167 (0.070)	-0.209 (0.045)	-0.337 (0.059)
Sequential Elimination $\times$ IQ	0.080 (0.101)	-0.037 (0.074)	-0.073 (0.141)	-0.050 (0.099)	-0.004 (0.062)	0.063 (0.087)
Minimum-Time Procedure $\times$ IQ	-0.035 (0.097)	0.104 (0.075)	0.437 (0.137)	-0.068 (0.095)	0.144 (0.065)	0.317 (0.100)
Inattention (ASRS)	-0.024 (0.035)	-0.001 (0.025)	0.008 (0.047)	-0.006 (0.035)	-0.016 (0.023)	-0.001 (0.036)
Selective attention	0.007 (0.028)	-0.006 (0.020)	-0.015 (0.035)	0.010 (0.027)	-0.030 (0.017)	-0.024 (0.021)
Working memory	0.091 (0.038)	-0.044 (0.027)	-0.073 (0.050)	-0.006 (0.036)	-0.006 (0.022)	-0.027 (0.035)
Attitude toward inconsistency	-0.027 (0.030)	0.005 (0.022)	-0.034 (0.038)	-0.019 (0.029)	0.015 (0.019)	0.026 (0.031)
Age	0.011 (0.007)	-0.009 (0.005)	-0.017 (0.010)	0.013 (0.007)	-0.010 (0.004)	-0.008 (0.007)
Female	-0.143 (0.172)	0.148 (0.131)	0.022 (0.244)	-0.094 (0.167)	-0.000 (0.113)	0.097 (0.181)
Education	0.107 (0.075)	-0.091 (0.057)	-0.071 (0.111)	0.042 (0.075)	-0.024 (0.048)	-0.015 (0.079)
Constant	-1.217 (0.798)	1.203 (0.538)	5.363 (1.039)	-1.828 (0.771)	2.353 (0.476)	5.761 (0.679)
Log α		-0.026 (0.188)	2.113 (0.078)		-0.017 (0.122)	1.723 (0.067)
Log-likelihood	-396.372	-663.528	-1239.567	-412.834	-937.810	-2016.481
Observations	616	616	616	616	616	616

Note: This table reports the regression coefficients for the effects of Sequential Elimination and the Minimum-Time Procedure on economic rationality relative to the Direct Procedure, estimated from models with treatment–IQ interactions. Columns 1 and 4 present coefficients from logistic regressions, while the remaining columns present those from negative binomial regressions. Robust standard errors are reported in parentheses.

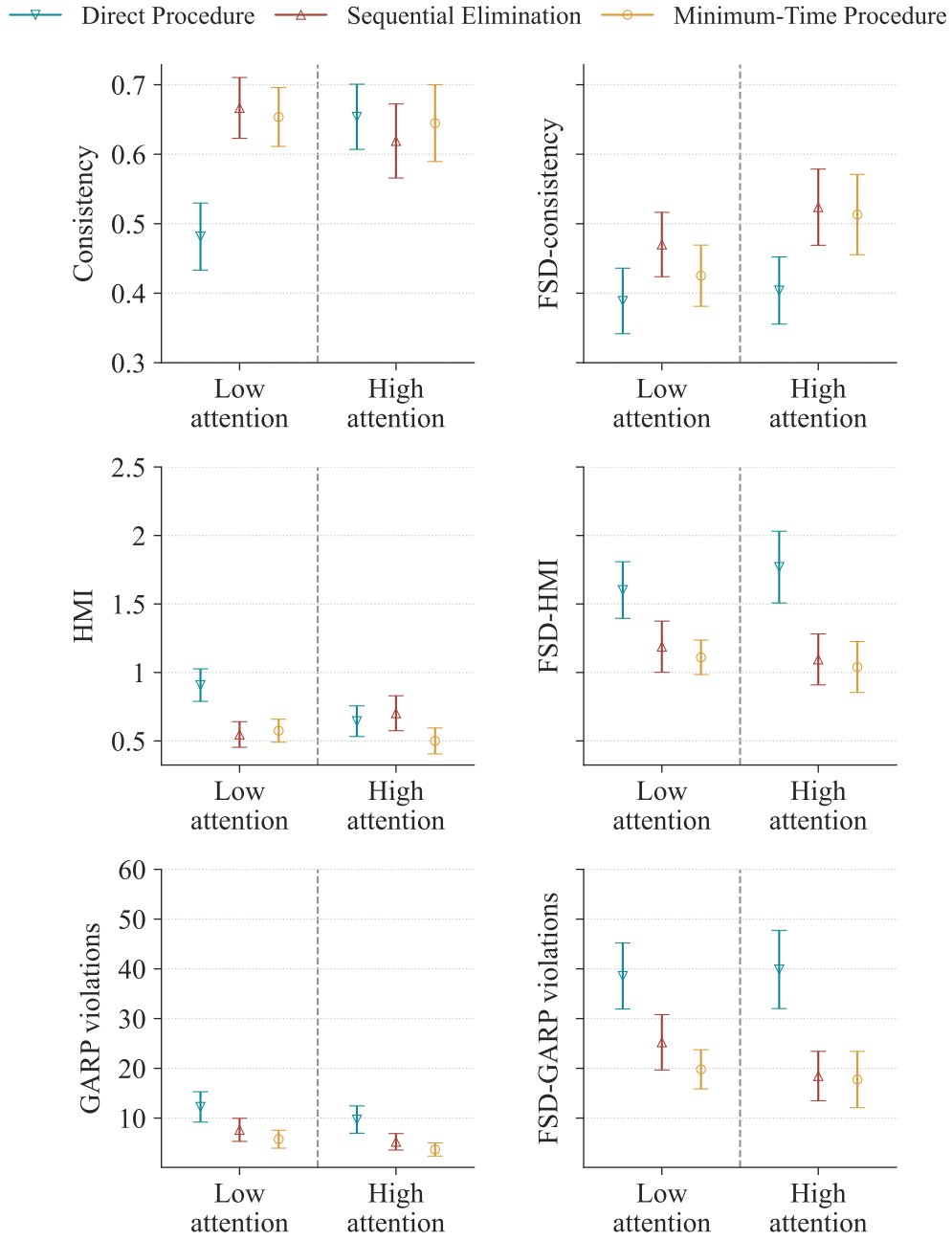


FIGURE D.3. Economic rationality across treatments by attention group. Mean values of each rationality measure are shown for low-attention participants and high-attention participants in each treatment, with error bars representing standard errors.

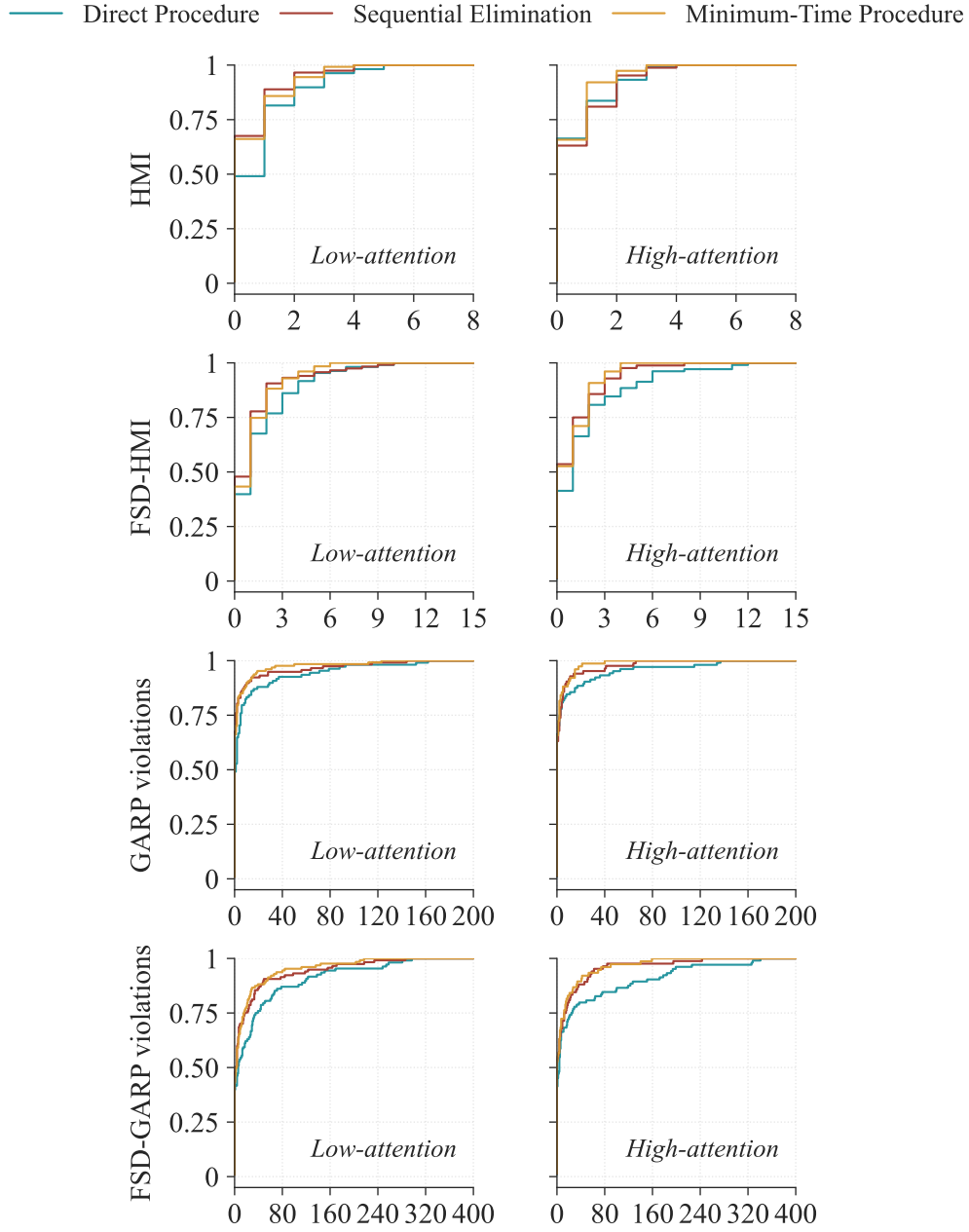


FIGURE D.4. Empirical cumulative distributions of rationality measures across treatments and attention groups.

TABLE D.5. Pairwise treatment comparisons by attention group across rationality measures.

Measure	SE vs. DP	SE vs. MTP	MTP vs. DP
<i>Panel A: Low-attention participants</i>			
Consistency	0.185 (0.005)	0.013 (0.829)	0.172 (0.008)
HMI	-0.360 (0.005)	-0.028 (0.736)	-0.333 (0.013)
GARP violations	-4.608 (0.003)	1.877 (0.810)	-6.485 (0.005)
FSD-consistency	0.081 (0.219)	0.045 (0.481)	0.036 (0.573)
FSD-HMI	-0.414 (0.074)	0.078 (0.498)	-0.492 (0.206)
FSD-GARP violations	-13.316 (0.047)	5.436 (0.854)	-18.752 (0.043)
<i>Panel B: High-attention participants</i>			
Consistency	-0.035 (0.622)	-0.026 (0.737)	-0.009 (0.899)
HMI	0.058 (0.638)	0.202 (0.445)	-0.144 (0.782)
GARP violations	-4.464 (0.882)	1.554 (0.616)	-6.018 (0.729)
FSD-consistency	0.120 (0.101)	0.011 (0.893)	0.109 (0.145)
FSD-HMI	-0.674 (0.078)	0.056 (0.935)	-0.730 (0.110)
FSD-GARP violations	-21.423 (0.101)	0.702 (0.940)	-22.125 (0.094)

Note: The table presents the differences in means for pairwise treatment comparisons across rationality measures among low-attention participants and high-attention participants. The corresponding p-values are reported in parentheses beside each difference. SE = Sequential Elimination, DP = Direct Procedure, MTP = Minimum-Time Procedure. Binary measures use chi-square tests; count measures use Mann–Whitney U tests.

TABLE D.6. Regression results on the effects of choice procedures on economic rationality with treatment-by-high-attention interactions.

	Consistency (1)	HMI (2)	GARP violations (3)	FSD-compliant		
				Consistency (4)	HMI (5)	GARP violations (6)
Sequential Elimination	0.839 (0.279)	-0.576 (0.204)	-0.532 (0.410)	0.385 (0.275)	-0.376 (0.181)	-0.535 (0.272)
Minimum-Time Procedure	0.713 (0.280)	-0.417 (0.200)	-0.555 (0.423)	0.117 (0.273)	-0.293 (0.179)	-0.397 (0.312)
High-attention	0.708 (0.289)	-0.357 (0.208)	-0.279 (0.375)	0.034 (0.281)	0.043 (0.177)	-0.083 (0.265)
Sequential Elimination $\times$ high-attention	-1.014 (0.418)	0.725 (0.316)	-0.050 (0.577)	0.122 (0.408)	0.007 (0.270)	-0.003 (0.434)
Minimum-Time Procedure $\times$ high-attention	-0.735 (0.426)	0.184 (0.321)	-0.125 (0.674)	0.371 (0.410)	-0.208 (0.270)	-0.234 (0.494)
IQ	0.076 (0.043)	-0.105 (0.033)	-0.095 (0.064)	0.128 (0.041)	-0.161 (0.028)	-0.194 (0.045)
Selective attention	0.009 (0.028)	-0.007 (0.020)	0.005 (0.035)	0.010 (0.027)	-0.031 (0.017)	-0.027 (0.023)
Working memory	0.091 (0.038)	-0.047 (0.027)	-0.053 (0.051)	-0.007 (0.036)	-0.002 (0.023)	-0.011 (0.037)
Attitude toward inconsistency	-0.025 (0.030)	0.008 (0.022)	-0.043 (0.039)	-0.023 (0.029)	0.019 (0.019)	0.015 (0.031)
Age	0.011 (0.007)	-0.009 (0.005)	-0.016 (0.010)	0.012 (0.007)	-0.008 (0.004)	-0.008 (0.007)
Female	-0.120 (0.172)	0.141 (0.131)	0.207 (0.259)	-0.109 (0.167)	0.010 (0.113)	0.159 (0.193)
Education	0.109 (0.076)	-0.095 (0.056)	-0.142 (0.110)	0.042 (0.075)	-0.026 (0.048)	-0.031 (0.078)
Constant	-1.746 (0.774)	1.306 (0.522)	4.320 (0.992)	-1.636 (0.728)	2.012 (0.449)	5.217 (0.696)
Log α		-0.035 (0.189)	2.147 (0.078)		0.007 (0.121)	1.740 (0.067)
Log-likelihood	-393.657	-662.357	-1245.354	-412.059	-941.070	-2020.691
Observations	616	616	616	616	616	616

Note: This table reports the regression coefficients for the effects of Sequential Elimination and the Minimum-Time Procedure on economic rationality relative to the Direct Procedure, estimated from models with treatment–High-attention interactions. Columns 1 and 4 present coefficients from logistic regressions, while the remaining columns present those from negative binomial regressions. Robust standard errors are reported in parentheses.

TABLE D.7. Comparison of economic rationality between low-attention and low-IQ participants.

	Direct Procedure	Sequential Elimination	Minimum-Time Procedure	Overall
Consistency	-0.019 (0.781)	0.012 (0.844)	0.046 (0.467)	0.021 (0.580)
HMI	-0.093 (0.788)	-0.145 (0.630)	-0.061 (0.497)	-0.116 (0.329)
GARP violations	-1.776 (0.843)	0.062 (0.747)	1.606 (0.604)	-0.390 (0.459)
FSD-consistency	0.067 (0.294)	-0.016 (0.812)	0.023 (0.718)	0.028 (0.452)
FSD-HMI	-0.644 (0.084)	-0.204 (0.775)	-0.142 (0.436)	-0.364 (0.070)
FSD-GARP violations	-16.572 (0.138)	-2.321 (0.992)	-2.524 (0.621)	-8.308 (0.190)
Observations (low-attention/low-IQ)	108/118	117/107	127/107	352/332

Note: This table compares economic rationality between low-attention and low-IQ participants within each treatment and in the overall sample. Mean differences (low-attention minus low-IQ) are reported with p -values in parentheses. These p -values are derived from chi-square tests for choice consistency and FSD-consistency, and Mann–Whitney U tests for the remaining measures.

TABLE D.8. Average marginal effects of choice procedures on economic rationality by ASRS group.

				FSD-compliant		
	Consistency	HMI	GARP			
	(1)	(2)	violations	Consistency	HMI	GARP
			(3)	(4)	(5)	violations
	(6)					
<i>Sequential Elimination</i>						
High-ASRS participants	0.166	-0.370	-5.687	0.090	-0.533	-17.937
	(0.062)	(0.145)	(3.630)	(0.062)	(0.244)	(7.826)
Low-ASRS participants	-0.014	0.074	-3.196	0.143	-0.498	-11.137
	(0.074)	(0.169)	(2.906)	(0.077)	(0.296)	(8.967)
Overall	0.096	-0.196	-4.712	0.111	-0.520	-15.276
	(0.048)	(0.112)	(2.469)	(0.048)	(0.191)	(5.898)
<i>Minimum-Time Procedure</i>						
High-ASRS participants	0.176	-0.365	-6.460	0.059	-0.568	-16.827
	(0.063)	(0.150)	(3.517)	(0.062)	(0.246)	(8.996)
Low-ASRS participants	-0.034	-0.042	-2.443	0.072	-0.436	-9.303
	(0.072)	(0.135)	(3.260)	(0.075)	(0.292)	(8.945)
Overall	0.094	-0.239	-4.888	0.064	-0.516	-13.884
	(0.048)	(0.106)	(2.459)	(0.048)	(0.187)	(6.320)
Observations	616	616	616	616	616	616

Note: This table reports average marginal effects of Sequential Elimination and the Minimum-Time Procedure relative to the Direct Procedure by ASRS group, estimated from models with treatment-by-low-ASRS interactions. Participants with total ASRS scores below the sample median are classified as low-ASRS, while those with scores at or above the median are classified as high-ASRS. Columns 1 and 4 present average marginal effects from logistic regressions, while the remaining columns present those from negative binomial regressions. All specifications control for IQ and selective attention, working memory, attitude toward inconsistency, age, gender, education. Robust standard errors are reported in parentheses.

TABLE D.9. Mean response time and its components by treatment and IQ group.

IQ group	Direct Procedure	Sequential Elimination	Minimum-Time Procedure
<i>Panel A: Low-IQ participants</i>			
Response time	20.673 (0.372)	23.871 (0.589)	40.847 (0.402)
Time to last click	18.607 (0.371)	22.379 (0.592)	15.279 (0.227)
Time after last click	2.066 (0.091)	1.492 (0.077)	25.567 (0.449)
Observations	2832	2568	2568
<i>Panel B: High-IQ participants</i>			
Response time	24.520 (0.607)	29.296 (0.726)	40.112 (0.480)
Time to last click	22.003 (0.599)	27.514 (0.696)	17.159 (0.268)
Time after last click	2.517 (0.077)	1.782 (0.211)	22.953 (0.528)
Observations	2256	2256	2304
<i>Panel C: Overall</i>			
Response time	22.379 (0.340)	26.408 (0.464)	40.499 (0.311)
Time to last click	20.113 (0.337)	24.780 (0.454)	16.168 (0.175)
Time after last click	2.266 (0.061)	1.628 (0.107)	24.331 (0.345)
Observations	5088	4824	4872

Note: This table reports treatment-specific mean timing metrics per decision problem across IQ groups. Response time is measured as the seconds elapsed from when a participant encounters a problem until they submit a choice. Time to last click is measured as the seconds elapsed from encountering a problem until the final click on any option. Time after last click is measured as the seconds elapsed from the final click on any option until submission. Observations are at the participant–problem level. Standard errors are reported in parentheses.

TABLE D.10. Regression results on the effects of choice procedures on response time and its components.

	Response time (1)	Time to last click (2)	Time after last click (3)
Sequential Elimination	3.180 (1.668)	3.819 (1.651)	-0.639 (0.248)
Minimum-Time Procedure	20.198 (1.187)	-3.244 (1.076)	23.442 (0.926)
High-IQ	3.233 (1.547)	2.562 (1.484)	0.671 (0.287)
Sequential Elimination $\times$ high-IQ	2.242 (2.678)	2.725 (2.641)	-0.484 (0.398)
Minimum-Time Procedure $\times$ high-IQ	-4.443 (1.860)	-1.342 (1.742)	-3.101 (1.383)
Controls	Yes	Yes	Yes
Decision problem fixed effects	Yes	Yes	Yes
Observations	14784	14784	14784

Note: This table reports linear regression results on how relevant time metrics vary across choice procedures. Response time is measured in seconds from encountering a decision problem until final choice submission. Time to last click is measured in seconds from encountering a decision problem until the last interaction with any option. Time after last click is measured in seconds from the last interaction with any option until final submission. Observations are at the participant–problem level. Controls include ASRS inattention, selective attention, working memory, attitude toward inconsistency, age, gender, education, and decision order. All specifications include a constant. Standard errors clustered at the individual level are reported in parentheses.

TABLE D.11. Counterfactual response time and rationality for low-IQ participants under the Direct Procedure

	Consistency (1)	HMI (2)	GARP violations (3)	FSD-compliant		
				Consistency (4)	HMI (5)	GARP violations (6)
Prediction at observed response times	0.500 (0.046)	1.000 (0.123)	14.780 (3.335)	0.322 (0.043)	2.232 (0.246)	57.232 (8.530)
Prediction at counterfactual response times	0.502 (0.048)	0.983 (0.126)	14.122 (3.212)	0.324 (0.045)	2.163 (0.246)	55.483 (8.594)
Counterfactual change	0.002 (0.015)	−0.017 (0.026)	−0.658 (0.335)	0.002 (0.013)	−0.069 (0.037)	−1.749 (1.215)
Implied response-time increase (s)	228.487	58.999	32.743	268.243	33.450	51.306

Note: This table reports counterfactual analyses for low-IQ participants under the Direct Procedure. The underlying models are logistic for choice consistency and FSD-consistency and negative binomial for the remaining outcomes, with a triple interaction among response time, treatment, and IQ group. Predictions are computed using observed response times and, for the counterfactual, the empirical response-time distribution of low-IQ participants under Sequential Elimination, holding all other covariates at their observed values. Counterfactual change is the difference between the two predictions. The implied response-time increase is the additional response time implied by the estimated models for each participant's predicted outcome to match the observed mean among low-IQ participants under Sequential Elimination. All models control for ASRS inattention, selective attention, working memory, attitude toward inconsistency, age, gender, education, and a constant. All reported point estimates are means, with robust standard errors reported in parentheses.

TABLE D.12. Average marginal effects of response time on economic rationality by treatment and IQ group.

	Consistency (1)	HMI (2)	GARP violations (3)	FSD-compliant		
				Consistency (4)	HMI (5)	GARP violations (6)
<i>Under the Direct Procedure</i>						
Low-IQ participants	0.001 (0.005)	-0.006 (0.010)	-0.269 (0.186)	0.001 (0.004)	-0.030 (0.021)	-0.791 (0.724)
High-IQ participants	0.003 (0.005)	-0.007 (0.010)	-0.047 (0.127)	0.003 (0.005)	0.000 (0.014)	0.207 (0.299)
<i>Under Sequential Elimination</i>						
Low-IQ participants	-0.002 (0.003)	-0.005 (0.008)	-0.272 (0.156)	0.001 (0.003)	-0.016 (0.014)	-0.684 (0.465)
High-IQ participants	0.006 (0.004)	-0.016 (0.008)	-0.193 (0.128)	0.008 (0.003)	-0.032 (0.012)	-0.323 (0.252)
<i>Under the Minimum-Time Procedure</i>						
Low-IQ participants	0.004 (0.007)	-0.003 (0.015)	-0.004 (0.133)	0.014 (0.007)	-0.035 (0.027)	-0.175 (0.717)
High-IQ participants	-0.006 (0.006)	0.002 (0.006)	-0.077 (0.122)	-0.008 (0.007)	-0.008 (0.011)	-0.108 (0.264)

Note: This table reports the average marginal effects of response time on economic rationality, conditional on IQ group and treatment, based on specifications including a triple interaction. Response time is averaged across decision problems at the individual level. Columns 1 and 4 present average marginal effects from logistic regressions, while the remaining columns present those from negative binomial regressions. All specifications include controls for ASRS inattention, selective attention, working memory, attitude toward inconsistency, age, gender, education, and a constant. Robust standard errors are reported in parentheses.

TABLE D.13. Average marginal effects of response time components and choice procedures on economic rationality.

	Consistency (1)	HMI (2)	GARP violations (3)	FSD-compliant		
				Consistency (4)	HMI (5)	GARP violations (6)
Time to last click	0.001 (0.002)	-0.008 (0.004)	-0.196 (0.078)	0.002 (0.002)	-0.018 (0.008)	-0.296 (0.232)
Time after last click	0.004 (0.004)	-0.013 (0.009)	-0.127 (0.120)	0.009 (0.005)	-0.034 (0.014)	-0.192 (0.385)
<i>Sequential Elimination</i>						
Low-IQ participants	0.139 (0.064)	-0.240 (0.155)	-6.809 (3.613)	0.160 (0.065)	-0.660 (0.283)	-29.217 (10.820)
High-IQ participants	0.018 (0.068)	-0.005 (0.113)	-2.201 (2.914)	0.026 (0.070)	-0.104 (0.182)	-3.034 (6.588)
Overall	0.083 (0.047)	-0.132 (0.100)	-4.685 (2.364)	0.098 (0.048)	-0.403 (0.176)	-17.146 (6.617)
<i>Minimum-Time Procedure</i>						
Low-IQ participants	0.027 (0.128)	-0.009 (0.317)	-8.244 (4.509)	-0.116 (0.104)	-0.004 (0.511)	-29.831 (15.332)
High-IQ participants	-0.012 (0.103)	0.136 (0.209)	0.522 (4.599)	-0.146 (0.104)	0.414 (0.345)	-2.191 (9.015)
Overall	0.009 (0.106)	0.058 (0.243)	-4.203 (3.859)	-0.130 (0.094)	0.189 (0.389)	-17.087 (10.899)

Note: This table reports the average marginal effects of time to last click, time after last click, Sequential Elimination, and the Minimum-Time Procedure on economic rationality. Both time metrics are averaged across decision problems at the individual level. Treatment effects are reported by IQ group. Columns 1 and 4 present average marginal effects from logistic regressions, while the remaining columns present those from negative binomial regressions. All specifications include controls for ASRS inattention, selective attention, working memory, attitude toward inconsistency, age, gender, education, and a constant. Robust standard errors are reported in parentheses.

TABLE D.14. Descriptive patterns of decision sequences by treatment and IQ group.

	Mean position in decision sequence			Spearman rank correlation
	FSD-dominated	Boundary	Middle	
	(1)	(2)	(3)	
<i>Direct Procedure</i>				
Low-IQ participants	6.107 (0.035)	6.146 (0.043)	5.901 (0.061)	0.700 (0.011)
High-IQ participants	6.124 (0.039)	6.092 (0.045)	5.921 (0.066)	0.769 (0.012)
Overall	6.115 (0.026)	6.122 (0.031)	5.910 (0.045)	0.730 (0.008)
<i>Sequential Elimination</i>				
Low-IQ participants	4.922 (0.035)	4.430 (0.051)	8.151 (0.063)	0.097 (0.011)
High-IQ participants	4.676 (0.035)	3.787 (0.053)	8.049 (0.065)	0.071 (0.011)
Overall	4.807 (0.025)	4.129 (0.037)	8.103 (0.045)	0.085 (0.008)

Note: This table reports treatment-specific characteristics of decision sequences by IQ group. Columns 1–3 present the mean position of FSD-dominated, boundary, and middle options in the decision sequence. Under the Sequential Elimination procedure, this corresponds to the elimination order; under the Direct Procedure, it corresponds to the examination order. Column 4 displays the Spearman correlation between the decision sequence and the presentation sequence, with lower values indicating weaker alignment with the presentation order. The Minimum-Time Procedure does not generate decision sequences and is therefore excluded. Observations are at the participant–problem level. Standard errors are reported in parentheses.

TABLE D.15. Treatment differences in elimination and examination sequences under Sequential Elimination and the Direct Procedure.

	Mean position in decision sequence			Spearman correlation
	FSD-dominated (1)	Boundary (2)	Middle (3)	
Sequential Elimination	-1.167 (0.086)	-1.727 (0.206)	2.295 (0.222)	-0.599 (0.051)
High-IQ	0.055 (0.076)	-0.055 (0.116)	0.111 (0.138)	0.076 (0.065)
Sequential Elimination $\times$ high-IQ	-0.288 (0.122)	-0.572 (0.296)	-0.175 (0.306)	-0.098 (0.075)
Controls	Yes	Yes	Yes	Yes
Decision problem fixed effects	Yes	Yes	Yes	Yes
Observations	9912	9912	9912	9912

Note: This table reports the linear regression results on the treatment differences in elimination and examination sequences under Sequential Elimination and the Direct Procedure. Columns 1–3 present estimates with the mean sequence position of FSD-dominated, boundary, and middle options as the dependent variable, respectively. Column 4 presents estimates with the Spearman correlation between decision sequences and the presentation order as the dependent variable. Observations are at the participant–problem level. Controls include ASRS inattention, selective attention, working memory, attitude toward inconsistency, age, gender, education, and decision order. All specifications include a constant. Standard errors clustered at the individual level are reported in parentheses.

TABLE D.16. Descriptive patterns of final choices by treatment and IQ group.

	FSD-dominated (1)	Boundary (2)	Middle (3)	Presentation position (4)	Mirrored options chosen (5)
<i>Direct Procedure</i>					
Low-IQ participants	0.055 (0.004)	0.117 (0.006)	0.378 (0.009)	6.078 (0.059)	0.440 (0.014)
High-IQ participants	0.020 (0.003)	0.131 (0.007)	0.370 (0.010)	5.900 (0.067)	0.505 (0.016)
Overall	0.039 (0.003)	0.123 (0.005)	0.374 (0.007)	5.999 (0.045)	0.469 (0.010)
<i>Sequential Elimination</i>					
Low-IQ participants	0.035 (0.004)	0.183 (0.008)	0.365 (0.010)	5.982 (0.062)	0.517 (0.015)
High-IQ participants	0.016 (0.003)	0.135 (0.007)	0.316 (0.010)	6.000 (0.068)	0.484 (0.016)
Overall	0.026 (0.002)	0.161 (0.005)	0.342 (0.007)	5.991 (0.046)	0.502 (0.011)
<i>Minimum-Time Procedure</i>					
Low-IQ participants	0.025 (0.003)	0.123 (0.006)	0.322 (0.009)	5.979 (0.063)	0.449 (0.015)
High-IQ participants	0.016 (0.003)	0.200 (0.008)	0.283 (0.009)	5.967 (0.066)	0.508 (0.015)
Overall	0.021 (0.002)	0.159 (0.005)	0.304 (0.007)	5.973 (0.046)	0.476 (0.011)

Note: This table reports mean frequencies of final choices by treatment and IQ group. Columns 1–3 present the frequencies of selecting FSD-dominated, boundary, and middle options, respectively. Column 4 displays the mean presentation position of the chosen option, with 6 corresponding to the midpoint of the 11-option list. Column 5 shows the frequency of selecting mirrored options in eleven symmetrically equivalent pairs of decision problems. Observations are at the participant–problem level in Columns 1–4 and the participant–pair level in Column 5. Standard errors are reported in parentheses.

TABLE D.17. Determinants of choice of Sequential Elimination.

	Choice of Sequential Elimination				
	(1)	(2)	(3)	(4)	(5)
High-IQ	-1.312 (0.588)			-1.060 (0.628)	
IQ		-0.286 (0.121)			-0.254 (0.133)
Selective attention	-0.107 (0.095)	-0.124 (0.098)		-0.130 (0.102)	-0.145 (0.104)
Working Memory	-0.210 (0.112)	-0.194 (0.112)		-0.239 (0.134)	-0.228 (0.130)
Education			0.567 (0.285)	0.627 (0.278)	0.663 (0.279)
Age				-0.032 (0.044)	-0.024 (0.043)
Female				0.668 (0.585)	0.684 (0.587)
Attitude toward inconsistency				0.038 (0.123)	0.020 (0.123)
Constant	4.692 (1.850)	5.637 (1.899)	-0.313 (0.504)	4.381 (2.315)	5.093 (2.344)
Log-likelihood	-41.147	-40.471	-45.901	-38.468	-37.630
Observations	75	75	75	75	75

Note: This table reports logistic regression coefficients for the choice of Sequential Elimination versus the Direct Procedure in the pilot treatment, where participants choose between the two procedures. The dependent variable equals 1 if Sequential Elimination is chosen and 0 otherwise. Robust standard errors are reported in parentheses.

TABLE D.18. Procedural preferences and economic rationality.

	Sequential Elimination Chosen vs. assigned		Direct Procedure Chosen vs. assigned	
	Consistency	HMI	Consistency	HMI
	(1)	(2)	(3)	(4)
<i>Panel A: Average marginal effects of choosing each procedure</i>				
Low-IQ participants	0.041 (0.109)	0.039 (0.260)	0.178 (0.211)	-0.312 (0.364)
High-IQ participants	-0.013 (0.134)	0.057 (0.213)	-0.176 (0.138)	0.446 (0.237)
Overall	0.019 (0.085)	0.046 (0.175)	-0.032 (0.121)	0.139 (0.209)
<i>Panel B: Average predictions for assigned low-IQ participants</i>				
Prediction under assignment	0.675 (0.077)	0.569 (0.157)	0.412 (0.085)	0.941 (0.164)
Prediction if chosen	0.716 (0.079)	0.607 (0.202)	0.589 (0.190)	0.625 (0.326)
Counterfactual change	0.041 (0.109)	0.038 (0.253)	0.177 (0.211)	-0.316 (0.365)
Low-IQ assigned participants	40	40	34	34
Observations	122	122	101	101

Note: This table reports results from the pilot treatment, in which participants could choose between Sequential Elimination and the Direct Procedure. It compares economic rationality for participants who choose each procedure with that of participants randomly assigned to the same procedure. Columns 1–2 concern Sequential Elimination and Columns 3–4 the Direct Procedure. Panel A reports average marginal effects of choosing rather than being assigned to each procedure, separately by IQ group. Panel B reports model-based counterfactual predictions for assigned low-IQ participants under assignment and counterfactually as if they had chosen the same procedure, holding other covariates at their observed values. Counterfactual change is the difference between the two predictions. Consistency is estimated by logit and HMI by negative binomial regression. All specifications control for selective attention, working memory, attitude toward inconsistency, age, gender, education and a constant. Robust standard errors are reported in parentheses.